TITLE

AFFORDABLE HYPERSPECTRAL IMAGING SYSTEM FOR IN-LINE FOREIGN BODY DETECTION AND CHEMICAL COMPOSITION ANALYSIS DURING FOOD PROCESSING

ACRONYM

HYPERA

DISRUPTIVE DETECTION TECHNOLOGY

PARTICIPANT LIST

Participant no.	Participant organisation name	Country
1 (Coordinator)	INNOVACIO I RECERCA INDUSTRIAL I SOSTENIBLE (IRIS)	Spain

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1. EXCELLENCE

1.1 OBJECTIVES

SUMMARY

Foreign body presence in foods is among the biggest source of complaint and prosecution in the food industry[1]. To the food processor or manufacturer, foreign body contamination can be costly as a result of product recalls, legal expenses and ultimately, the costs of lost reputation and consumer confidence. There is a growing demand for affordable and reliable foreign body detection technology that is fast enough for its in-process use in modern food processing plants. Through this SME Instrument project, we, a Spanish advanced engineering company specialising in on-line process monitoring and control systems, aim to accelerate the final stages of development and gear up for the commercialisation of a new **hyperspectral imaging (HSI) system** (HYPERA®) for the cost effective on-line detection of foreign bodies and chemical composition analysis that will be targeted to the global food industry.

Thanks to its novel and disruptive combination of Shortwave Infrared Hyperspectral Imaging (SWIR-HSI) in reflectance with Near Infrared Multispectral Imaging (NIR-MSI) in reflectance/transmittance, the innovative HYPERA system (Figure 1), which has been awarded 'Seal of Excellence' certification from the European Commission (see Annex 1), will differentiate itself in the market for its high resolution, its ability to penetrate foodstuffs, its composition differentiation, its compact design and affordability, and its high speeds and versatility. The system will be deployed with its own software package that will contain specialised chemometric tools, to offer the food industry with a turnkey solution that is readily customisable for foreign body detection of a range of contaminants, in a wide variety of matrixes and suitable for integration in multiple modern food processing environments. Building on a prototype system (see figure 2 below), which has been validated to TRL6 as a forerunner to this project, over the course of the Phase 2 SME Instruments, four application cases will be developed in foreign body detection in pizzas, ready-to-eat (RTE) foods, processed fruit and vegetables, and cooked meats, whereby the imaging systems will be integrated into a rugged system hardware and the software and chemometrics will be further developed installed and trialled in processing lines in the industry in order to demonstrate their correct functioning in an operational environment at pre-commercial scale. Positive results from this industry demonstration work will enable us to translate the prototype to a first commercial system and to ensure that manufacturing issues solved) by the end of the project.

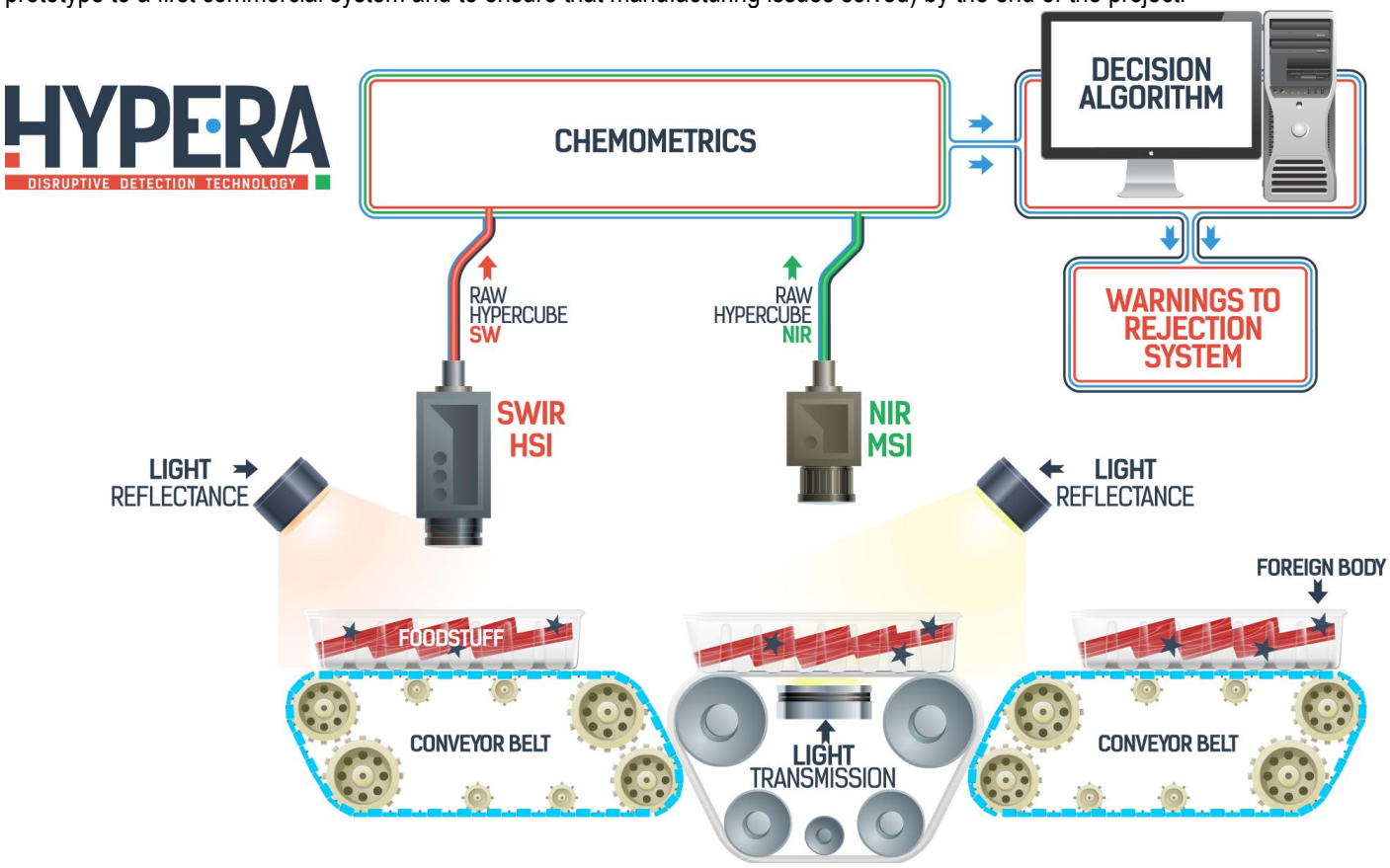


FIGURE 1- THE HYPERA® HSI FOREIGN BODY DETECTION SYSTEM

[1] Rotimi Toki (2016). Foreign Bodies in Foods - Effective Techniques for Prevention, Control and Detection. http://www.compliance4all.com/control/w_product/~product_id=500617LIVE

In the immediate post-project phase, we will fine tune as necessary to a full commercial application ensuring that the technology is available for consumers by January 2019. In parallel with the product and application development work, we will also prepare for the commercialisation phase to deliver on a business plan, whereby the commercial targets are to sell 500 units in the first 5 years of market entry, leading to a cumulative turnover of €46M for that period, and 42 new jobs being created in IRIS.

SPECIFIC OBJECTIVES OF THE PROJECT

With a view to having a near commercial HYPERA® systems for affordable, rapid and non-destructive foreign body detection in foods by the end of this Phase 2 SME Instrument project, the following objectives will be fulfilled during the 24-month workplan:

1. To develop the overall system specifications and to establish the KPIs for validating the system *(Linked to WP1)*

The user requirements of our collaborating early adopters[2] will be systematically established so that detailed system specifications can be drawn up: 1) CASA TARRADELLAS, a leading Spanish pizza manufacturer; 2) PLATOS TRADICIONALES, an important Spanish producer of Ready-to-Eat (RTE) foods such as lasagne, rice dishes and roast dishes (supplying MERCADONA, one of the leading retail chains in Spain); 3) MARÍN GIMENEZ an important Spanish fruit and vegetable processor; and 4) DAWN FARM FOODS from Ireland, a leading European supplier of cooked and fermented meat ingredients to the foodservice and manufacturing sectors, as well as a wider group of food processors. A key aspect of this process is the early implementation of a pilot test with existing prototype equipment to assist the identification of key performance goals. Foreign body detection success criteria should be carefully defined on the basis of considerations that take into account their probabilistic nature, which is governed by the Poisson distribution whereby foreign body occurrence is essentially a "rare event" if the production process has been designed following a HACCP analysis. ***Performance Indicator(s):*** By M3, the system specifications are defined and the KPIs for determining the success of the developed system are established.

2. To develop the chemometric predictive models for each application *(Linked to WP2)*

The aim is to use the New powerful data processing tools will be implemented based on the most recent advancements in chemometrics for the high speed and multifunctional chemical analysis of the data coming from the sensors and for minimizing the image distortion caused by scattering. In fact, derived from early developments for imaging through both irregular atmospheric paths and biological tissues (medical diagnosis), a variety of non-invasive techniques are currently available for overcoming blurred outputs by means of purely mathematical approaches. We will merge such available mathematical tools with chemometric routines in order to simultaneously attain spatial and chemical information regardless the scattering due to the nature of the medium. Using the feasibility prototype system in IRIS, initial tests with product samples sets from our four beta testing clients will be performed for obtaining test data for the development project and for model development. The models relating spectral data will be validated in the laboratory with selected chemical, physical and microbiological parameters in selected food matrixes and against available detection methods following traditional (such as artificial vision in the VIS range) and laboratory (i.e., destructive inspection scheme) methods. An initial set of lab-scale ready-to-use predictive chemometric models will be developed, validated, and pre-installed in the respective HYPERA units to be validated. Novel chemometric methods (MCR - multivariate curve resolution- and MLR –multivariate linear regression- improved with wavelengths selection pre-treatment –genetic algorithms- as well as robust classification methods like SVM –support vector machines-) will be implemented for improving accuracy in relation to standard methods like PLS or PLS-DA, for example. ***Performance indicator:*** By M7, digital files for the first versions of each model are ready for laboratory validation and by M9, optimised versions are delivered for coding into the system software.

3. To design and build the instrumentation hardware and software prototype *(Linked to WP3)*

The Shortwave Infrared Hyperspectral Imaging (SWIR-HSI) in reflectance with Near Infrared Multispectral Imaging (NIR-MSI) in reflectance/transmittance will be scaled up and combined testing performed to verify the effectiveness of the novel combination for offering penetration (NIR) in foodstuffs and for composition differentiation (SWIR) in the target application cases. The industrial HYPERA® system will be designed in keeping with industry specifications to deliver the following threshold requirements: a) operation at a speed of 1 m/sec; b) penetration at depths of 25 mm (depending on the nature of both foreign body and medium); c) achieving false negative detection rate of <5-10% of foreign bodies cases which cannot be detected by means of X-ray systems or inductive detectors d) ease of integration into existing processing lines. Five functional HSI prototypes will be designed and built (one each for integration into our four food processing end-users- CASA TARADELLAS, PLATOS TRADICIONALES, MARÍN GIMENEZ and DAWN FARM FOODS for industry trials)- and one for optimisatio work. Each will have extended infrared bands using novel detectors to acquire the most important object features, employing the latest technologies for compact hyperspectral imagers, and such that the cost of the instrument can be efficiently limited. The system control software will be developed, incorporating the chemometric models and the user-friendly User Interface. ***Performance Indicators:*** By M10, design details, system building instructions and code of the integrated system are ready. By M13, the hardware and software are ready for integration. By M16, the integrated system is tested and ready for installation in the industry. By M24 the system has is delivered to TRL8-9.

4. To integrate the system hardware and software and to perform pre-validation tests to ensure proper functioning of the hard- and software *(WP3)*

Performance indicator: By M12, tests with the integrated system confirm readiness for industry integration work to commence and the Factory Acceptance Test (FAT) according to ISO 2001:9008 is approved.

5. To install and trial the HYPERA® system in the food processing industry application cases *(Linked to WP4)*

The HYPERA system will be installed in: 1) a pizza processing line of CASA TARRADELLAS for the detection of plastic and other foreign

[2] See letters of support that have been provided by these companies- Annex 1 of this bid

bodies in pizza, detection of fragments of fruit seeds (kernels) and extrinsic foreign bodies (plastic, stones…); 2) a packing line of PLATOS TRADICIONALES industry to greatly facilitate quality control (which is currently performed by sampling of raw materials, X-Ray and metal detectors and tedious visual inspection); 3) in fresh cut fruit conveyor at MARÍN GIMENEZ; and, 4) in one of the cook lines at DAWN FARM FOODS, whereby they are mainly concerned with plastic detection. We will extensively test the prototypes and evaluate their performance during commercial processing for foreign body detection according to the pre-defined KPIs. Comparisons will be made with conventional quality control methods in use at each plants in order to industrially validate the system. Necessary modifications will be identified to improve the productised system***. Performance Indicators:*** By M14, the systems are successfully installed at the facilities of our 4 early adopters is obtained from each of the sites. By M24, the impacts and KPIs of having the systems installed and in operation for 10 months are quantified and compared vs. historian data.

6. **To demonstrate to food processors the ease of integration of the system, and its advantages for rapid, non-destructive foreign body detection during commercial processing** *(Linked to WP4)*

A public workshop will be organised by IRIS to showcase the system in operation targeting the Quality Managers from the food industry. Results of the long term operational tests at the industry test sites will be presented by our collaborating End Users. To facilitate demonstrations a pilot scale installation will be provided. ***Performance Indicators:*** By M24, the system is demonstrated to >150 food processing companies and actors from the value chain.

7. **To translate the validated prototype designs to a product through the creation of the manufacturing documentation package using the engineering information and specifications and to establish the manufacturing.** *(Linked to WP5)*

Standardisation and regulatory issues will be addressed to ensure compliance with production conditions. The aim is to reach Technology Readiness Level 8/9 and to arrive at MRL9-10, whereby the manufacturing infrastructure can be set up in order to be in a position to serve first orders by January 2019. ***Performance Indicators:*** By M19, the product manufacturing documentation packages are passed to purchasing and production engineering to enable manufacture. By M23, the first 5 product units pass the quality assurance tests.

8. **To gear up to the market** *(Linked to WP6)*

This will involved the establishment of the HYPERA Application Team, the training of the Sales Team, the finalization of the pricing structure and policy, and the commercial/distribution networks for delivering on the sales targets in our business plan. ***Performance Indicators:*** By M22 the Application Team is in place and trained, the first tier commercial network is in place and our pricing has been validated for timely progression to the Commercialisation Phase.

PROBLEM OR BUSINESS OPPORTUNITY

Food safety and quality control are essential parts of the food industry. EU has one of the highest food safety standards in the world – largely thanks to the solid set of EU legislation in place, which ensures that food is safe for consumers. Food processors are mandated by law to produce safe foods; EU legislation requires that food complies with various hygiene regulations specifying that food must be safe. All potential hazards must be eliminated or reduced to an acceptable level. A key tool to ensure the cross-border follow of information to swiftly react when risks to public health are detected in the food chain is RASFF – the Rapid Alert System for Food and Feed.

Foreign body detection in food products is a priority for every established food producer. Foreign bodies can be present in products from the field (stones, metal, insects, undesirable vegetable matter such as thorns or wood, dirt, or small animals), and can also result from processing and handling (bone, glass, metal, wood, nuts, bolts, screening, cloth, grease, paint chips, rust, and so on). While the unit operations in processing plants should be designed to remove or eliminate these physical hazards, despite the best efforts of producers, foreign bodies can be present in food that reaches the consumers, leading to injury (sometimes death, especially in infants), loss of loyalty, product rejections and recalls. Indeed, **foreign body contamination is recognised as one of the most common reasons for food product recalls[3].**

There has been a dramatic increase in the number of food recalls over the last few years, contributing factors including the increasingly global and complex food supply chain. Recent high profile and large scale recalls include the countless recalls as a result of the horsemeat scandal in Europe, the intentional shipment of salmonella contaminated peanut products by the Peanut Corporation of America, the recall of over a half billion fresh eggs from Wright County Egg and Hillandale Farms in the US, to mention but a few.

While food recalls are first and foremost a public health issue, they also pose significant economic impacts. **In fact, product recalls are considered one of the biggest threats to profitability for the food industry[4].** According to a joint industry study by the Food Marketing Institute and the Grocery Manufacturers' Association, the average cost of a recall to a food company has been estimated at $10M in direct costs[5], in addition to brand damage and lost sales (while this is a US study its results are transferable to the international context). For SMEs this could put them out of business. The costs for larger brands may be significantly higher based on the preliminary recall costs reported by firms of some recent recalls[6]. In short, the impacts can be devastating for food manufacturers and can cost them millions in damages and lost revenue, as well as severely damaging brand image that can take years to recover.

[3] Aoife A. Gowen, Colm P. O'Donnell (2013). Near infrared hyperspectral imaging for foreign body detection and identification in food processing. Spectroscopy Europe. VOL. 25 NO. 6 (2013). Pages 6-11
[4] Food Defense Whitepaper Series: Part Two. Recall: The Food Industry's Biggest Threat to Profitability. TYCO Integrated Security
[5] "Recall Execution Effectiveness: Collaborative Approaches to Improving Consumer Safety and Confidence," Deloitte on behalf of the Food Marketing Institute ("FMI"), the Grocery Manufacturers Association ("GMA"), and GS1 U.S., June 2010
[6] SAGE Food Safety Consultants

The industry's key focus is to **prevent these problems and incidences from ever occurring in the first place**. Consequently, manufacturers and producers are increasingly investing in identification methods of foreign bodies in order to protect their brands and for obtaining third-party certification to demonstrate their compliance with legal food safety requirements, to promote customer confidence and brand value.[7] Glass, metal and plastic are the most frequently cited foreign bodies in processed foods[3]. Currently, there are multiple types of foreign body detection systems available in the market ranging from simple to complex. Metal detectors are commonly implemented in food processing chains to prevent metal fragments occurring in finished products; however, these instruments are not capable of detecting glass, plastic and other contaminants in food. X-ray transmission radiography has long been used for online detection of foreign bodies in food products relying on the absorption contrast between the foreign body and food product[8]. While X-ray can detect a variety of foreign bodies such as glass, dense plastic, conglomerations, bones, and other objects with different densities, they cannot always detect very small contaminants such as hair, fine glass, or low-density contaminants such as cardboard, wood or insects[9]. Moreover X-ray is expensive and poses safety concerns.

HOW OUR SOLUTION SOLVES THE PROBLEM OR AVAILS OF THE BUSINESS OPPORTUNITY

In order to bridge a **market gap for improved, affordable, rapid, non-destructive techniques for foreign body detection and identification, and for real-time chemical composition analysis** in products in the food industry, IRIS is developing a hyperspectral imaging (HSI)- based system for foreign body detection in foods. **HSI** integrates both spectroscopic and computer vision techniques in one system for providing both spectral and spatial information simultaneously. This combination of imaging and spectroscopy enables enhanced non-destructive analysis of materials[3]. One major advantage of HSI over other methods of foreign body detection is the ability to detect, as well as classify, foreign bodies according to their spectral properties[3]. In short, if conventional spectroscopy is a technique to provide the answer to the question of "What", and conventional computer vision is a technique to provide the answer to the question of "Where", HSI provides the answer to the question of "Where Is What".

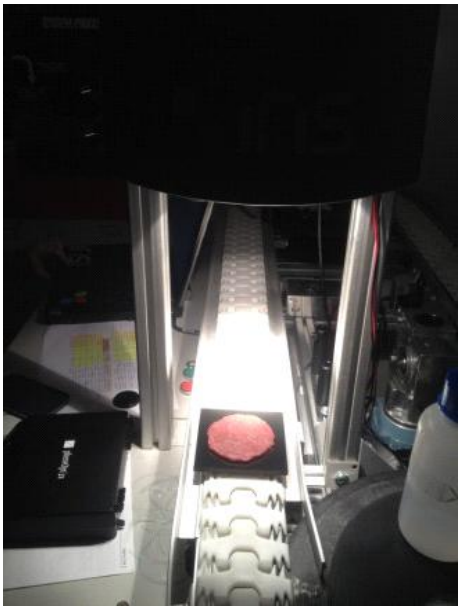


FIGURE 2- HYPERA PROTOTYPE INSTALLED IN A MINI PILOT LINE

When combined with chemometric (i.e. multivariate) analysis, HSI can be applied for prediction of the concentration and distribution of multiple components in a sample. This makes the technique attractive for many routine quality inspections within the food industry, such as body detection in food processes[3]. Recently, the technique has been widely studied for its applications in the food industry (discussed in more detail in Section 1.4) and positive results have been shown for example for the ability of HSI to identify glass, rubber, plastic, and other objects on the surface and subsurface of various food matrixes.

Numerous experimental and engineering research in the field of HSI have resulted in many successful applications in the food and pharmaceutical quality control. Despite striking advantages of HSI, such as its chemical-free, non-destructive detecting nature, together with the abundant information provided that enables multiple quality features to be examined[10], **moving this technology from the laboratories to the industry production lines poses a real challenge**. This is mainly due to the **low cost-efficiency of current HSI systems** and the **insufficient speed and accuracy of analysing complex information**. In particular, the rich information required for high accuracy, and the simplified design of the instrument to ensure high speeds and low cost, are currently contradictory industrial demands. Solutions are needed that displace this contradiction. This is the proposition of the HYPERA® system, whereby we bring a revolutionary new vision of real-time quality and safety assessment to the food and pharmaceutical industries. Our solution is based on a series of revolutionary design thinking concepts and developments (Table 1).

TABLE 1- REVOLUTIONARY NATURE OF THE HYPERA SYSTEM

1.	**"Specialised food assessment hyperspectral imager" vs. a "general hyperspectral imager for the food industry" that can perform a multifunctional and highly efficient inspection at significantly low cost**
A number of hyperspectral imagers for food inspection have been developed over the last 5-7 years (see Table 4 in Section 2). However these systems use the same principles and working modes as classical remote sensing hyperspectral systems, typically acquiring a lot of superfluous information, which occupies a large part of the system resources and reduces the working speed. Based on the in-depth analysis of the industrial demands (whereby for example we take into account the lower resolution requirements for food inspection vis-à-vis that offered by general	

[7] Frost & Sullivan, Breakthrough Technologies Enabling Food Safety.
[8] Mikkel Schou Nielsen, et. al. (2013). X-ray dark-field imaging for detection of foreign bodies in food. Food Control. Volume 30, Issue 2, April 2013, Pages 531–535
[9] Melissa Lind (2013). 5 Methods For Foreign Body Contamination Detection. July 30, 2014. Available at: http://www.foodonline.com/doc/five-methods-for-foreign-body-contamination-detection-0001
[10] Yuan-Yuan Pu, Yao-Ze Feng andDa-Wen Sun (2015). Recent Progress of Hyperspectral Imaging on Quality and Safety Inspection of Fruits and Vegetables: A Review. Comprehensive Reviews in Food Science and Food Safety. Volume 14, Issue 2, pages 176–188, March 2015. DOI: 10.1111/1541-4337.12123

hyperspectral imagers , and inspired by recent research results that have indicated that only a small number of spectral bands and potentially a special region of interest (ROI) will be enough for each function, employing the latest technologies for compact hyperspectral imagers we have designed and built a hyperspectral imaging system (currently at TRL6) based on an affordable and flexible design concept that has been conceived for its suitability for industrial lines.

2. Combination of chemometric tools and hyperspectral imager design for increased efficiency in terms of data acquisition and analysis to enable high processing speeds

Chemometrics is an essential tool for extracting useful information from massive spectral data of hyperspectral images, and there have been many applications of chemometrics to food and pharmaceutical products together with hyperspectral imagery. Unfortunately, however, each present chemometric method can only satisfy a few purposes with specific observing wavelengths, resulting in a low universality and efficiency of the system. Meanwhile, the analysis of huge amounts of data causes a low inspection speed that does not meet the demands of industry production lines. The combination of chemometric tools and hyperspectral imager design, however, is a potential way to significantly increase the efficiency of data acquisition and analysis by selecting useful spectral bands to read out according to the chemometric requirements. In doing so, the data volumes of both the instrument and the data processing system will be greatly reduced, resulting in shorter detecting periods making it possible to realise real-time assessment for production lines.

3. Obtaining more information by extending the observing spectral range

Extending the observing spectral range to infrared bands has been investigated as a potential way for obtaining more information. Over the last 3 years, visible and near-infrared (VNIR, 400–1000nm) as well as short-wave infrared (SWIR, 900–2300nm) HSI inspection systems have been active fields of research. These systems integrate VIS, NIR and SWIR ranges within the same hypercube (dataset); however, such ranges are not totally compatible in terms of penetration and composition. These existing multi-range systems are only suitable for some infrequent situations when a trade-off among the specific optical features of each spectral range can be achieved. HYPERA® on the other hand combines NIR and SWIR ranges in two solidary but independent modules that sum up the advantages of each region as each module is optimized in terms of lighting, signal gain and raw data processing methods. Moreover, the extended detecting spectral range will give the chemometric tools more available spectral bands to choose from and will allow the system to integrate a variety of chemometric methods and select different ones for various purposes and objects, thereby realising multifunctional inspection.

4. A powerful HSI software for acquiring hypercubes and visualizing foreign bodies or irregular spatial distribution of components and for improving the imaging capabilities and minimising the scattering effect

Seamless compatibility with the only chemometric suite in the market (SOLO + MIA, Eigenvector Research Inc) that includes tools for hyperspectral models. We will also incorpoporate MCR routines in Matlab. The compatibility with SOLO implies compatibility with Matlab and this converts our software in a semi-open tool, which permit limitless future applications that can be incorporated based on the latest achievements generated in the academic world.

In order to respond to a growing market demand for equipment that can assist food processors in guaranteeing the quality and safety of their foods and protecting their brands, this Innovation project will significantly advance the state-of-the-art by developing and bringing to market a **multifunctional, high speed HSI system suitable for quality control in production lines in the food and pharmaceutical industries, whereby the cost will be significantly lower and the performance will be higher than the commercially available systems**. As such, our project relates well with the SMEInst-07-2016-2017 work programme topic **'Stimulating the innovation potential of SMEs for sustainable and competitive agriculture, forestry, agri-food and bio-based sectors'**, whereby we will contribute to the supply of safe, healthy and high quality food, through the development and deployment of a resource-efficient eco-innovative food processing system. Moreover, HYPERA is in line with the wider objectives of H2020's Leadership in Enabling Technologies (LEIT) work programme, as it leverages photonics and ICT KETs (Key Enabling Technologies) to drive forward advanced processing technologies for improved industrial competitiveness. It is also in line with the Industry 4.0 paradigm, as it will contribute to food (and pharmaceutical) processors adopting data-driven production processes.

OBJECTIVES AND EXPECTED OUTCOME OF THE INNOVATION BUSINESS PROJECT

IRIS is positioning itself as an emerging player in the advanced monitoring and control space, whereby we have been offering bespoke turnkey on-line process monitoring solutions to a growing client base in the food, and more recently in the pharmaceutical, industries. From a business perspective, our goal through this project is to complete the development of a commercial HYPERA® HSI hardware and software system by the end of the Phase 2, to have the manufacturing set up with the first product units manufactured and quality approved, and to have finalised and validated our business model and pricing structure. The Phase 2 project enables us to accelerate this work so that we can be ready for market launch in Jan 2019; this acceleration will allow us to seize the market opportunity as early as possible and to be first to market with an affordable hyperspectral imaging system that is suited for in-line foreign body detection and chemical composition analysis in food processing. In parallel to the product development itself, we will also establish an Application Team (an internal team that will support sales and deployment activities: testing materials, performing detailed site visits, installing the systems, provide training and handling both site and remote customer support, etc.), and we will have our first tier international sales network established for Ireland & UK, Spain, Germany and Central Europe, the US and China. We will also work on commercialization planning for realising our business plan objectives of 500 units being sold (in Europe and worldwide) during the first 5 years of commercialization (rational provided in Section 2.2), which will translate into a cumulative turnover of €46 M (generated from 2019 to 2023) and will enable 42 new jobs to be created in IRIS over that same period.

1.2 RELATION TO THE WORK PROGRAMME

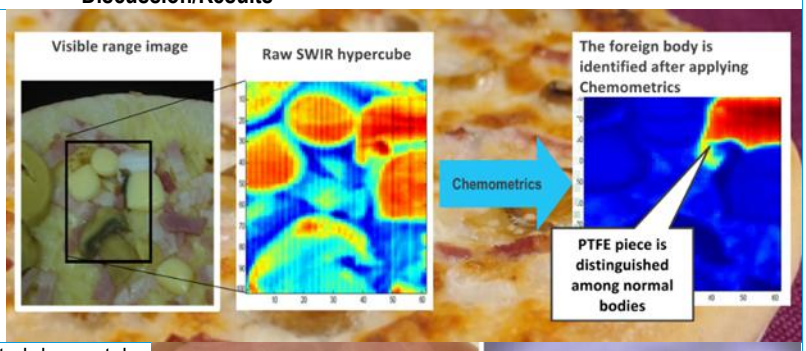
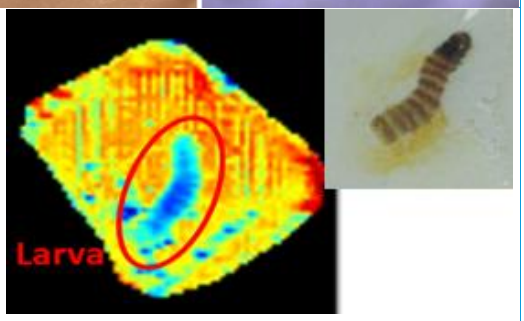
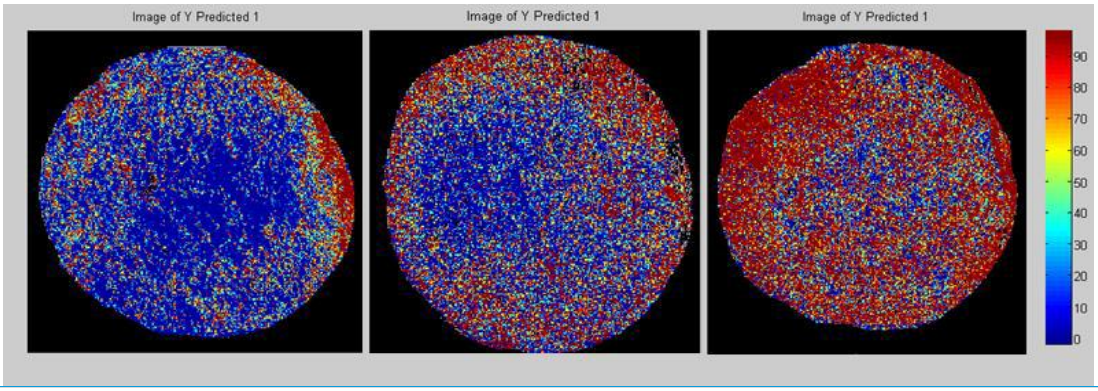
This proposal relates to ***SMEInst-07-2016-2017 work programme topic 'Stimulating the innovation potential of SMEs for sustainable and competitive agriculture, forestry, agri-food and bio-based sectors'.***

1.3 CONCEPT AND APPROACH

(A) CONCEPT

CURRENT STAGE OF DEVELOPMENT OF THE BUSINESS INNOVATION PROJECT

TABLE 2- SUMMARY OF APPLICATION TRIALS PERFORMED WITH THE HYPERA PROTOTYPE TO TRL6

Trial	Discussion/Results
Detection of Polytetrafluoroethylene (PTFE) pieces in pizza toppings	There is a risk of finding small (1-3 cm) irregular Teflon pieces that drop from grinders and that can be easily mistaken for white cheese cubes in the topping. Trials with samples from CASA TARRADELLAS using the HYPERA SWIR hyperspectral prototype verified the ability to easily detect Teflon pieces after applying chemometrics (see image). 
Detection of small metal bodies in between sliced ham layers	Small metal bodies cannot be easily detected by metal detectors based on RF due to the minimum size and mass limit of detection of such a technology. The significant penetration of NIR radiation allows such bodies to be distinguished as they are totally opaque and contrast against the surrounding homogeneous and transmissive milieu. The image shows a thin metal disk (coin) hidden in a 1-cm thick layered cooked ham. No chemometric pretreatment was required for unveiling the body because of the significant meat transparency at NIR range.
Detection of Anarsia lineatella in Prunus persica	The peach worm (peach twig borer, *Anarsia Lineatella Zeller*) is a major pest of stone fruits, such as peaches, nectarines and apricots. Quality control of processed peach pieces is currently carried out by means of human inspection. However, there are no relevant colour differences between peach worms and peach pieces, making difficult human-eye-based larvae detection. Typical pieces of peach –taken from the processing line- have been analysed with the HYPERA system: as peach worms differ in terms of chemical composition from the peach pieces, the peach worm can be easily detected. Fat, proteins and water are major components of worms, whereas peaches are mainly made of sugars, cellulose and water. 
Discrimination of beef vs. horsemeat and identification of the lowest percentage of minced meat adulteration that can be reliably detected.	HSI in the NIR region is adequate for the detection of meat adulteration with a precision of 8.8% in a range of 0 to 100% using a PLS model with a R2 of 0.94 and a RMSEP of 4.4. The HSI image prediction is able to detect meat adulteration as well as to study the sample heterogeneity and thus the meat mixture and the procedure elaboration effectiveness. Results show the promise of 

	establishing a rapid and non-invasive technique that can give rapid results. Meat image prediction. Image on the left corresponds to a 5% horsemeat, the middle image corresponds to a 50% of horsemeat and the image on the right side corresponds to a 95% horsemeat.

This action is built on an internal development programme undertaken by IRIS to combine two complementary features of near infrared spectroscopy according to the spectral range for in-depth penetration (NIR) and compositional information (SWIR). To date a prototype unit has been designed and built and integrated into a mini-pilot processing line (see Figure 2 above), and a proprietary Hyperspectral Imaging acquisition and chemometrics software has also been developed to beta level for acquiring hypercubes and visualizing foreign bodies or irregular spatial distribution of components and for improving the imaging capabilities and minimising the scattering effect (a screenshot of the main window can be seen in the image on the cover page and the improved setup for SWIR diffuse reflectance acquisition is also shown). This integrated system (HSI system combined with chemometrics) has been validated in conjunction with food industry partners for numerous applications using client samples, as is summarised in Table 2.

POSITIONING OF THE OF THE BUSINESS INNOVATION PROJECT

The starting point for the technological developments involved in HYPERA® are at sufficiently high Technology Readiness Levels (TRLs), i.e. TRL6/7, for their acceleration to the market and ensuring a swift time-to-commercialisation (by Jan 2019). Over the course of the Phase 2 SME Project, building on the existing prototype system that has been developed by IRIS (see figure 2 above), and already tested in intended food processing environments close to expected performance (TRL6), four applications will be developed in foreign body detection in pizzas, fruit, cooked meats, and Ready-to-Eat meals, and the systems will be installed, validated and demonstrated in the industry in order to demonstrate their correct operation in an operational environment at pre-commercial scale (TRL7). Positive results from this industry demonstration work will enable us to translate the prototype to a first commercial system and to ensure that manufacturing issues solved (TRL8) by the end of the project. In the 6-9 month project phase, we will fine tune as necessary to a full commercial application ensuring that the technology is available for consumers (TRL9) by Jan 2019. The sub-units of the HYPERA solution are each positioned with respect to their current TRLs and the expected TRLs by the end of this project (Table 3).

TABLE 3- TRL ADVANCEMENTS THROUGH THIS PROJECT

System sub-units	Current TRL	TRL at end of project
Shortwave Infrared Hyperspectral Imaging (SWIR-HSI) in reflectance	TRL6	TRL8/9
Near Infrared Multispectral Imaging (NIR-MSI) in reflectance/transmittance	TRL6	TRL8/9
Chemometric software	TRL7	TRL9

(B) METHODOLOGY

CONCEPT AND ACTIVITIES THAT WILL BE IMPLEMENTED DURING THE PROJECT

Our project concept is now to further develop the HYPERA® system hardware and software in order to advance the TRLs to 8/9- i.e., to complete the development of a system suitable for being successfully commercialised- and to demonstrate a system that is capable of tackling the widest set of typical foreign body detection challenges. In relation to our project approach to the work, our HYPERA project follows an **application-oriented product development approach**, which will involve a tight co-creation between the development team and our industry end-users (CASA TARRADELLAS, PLATOS TRADICIONALES, MARÍN GIMENEZ, DAWN FARM FOODS). These beta clients permit the project to use real applications as a basis for the development (a summary of the target applications is provided in Table 4).

TABLE 4- TARGET APPLICATIONS FOR THE PHASE 2 PROJECT

Company	Sector/Target Products	Target Foreign bodies	Installation
CASA TARRADELLAS	Pizza	Pieces of plastic from the packaging of topping components, wood, insects and PTFE pieces from cheese grinders in a variety of different toppings (more than 10 types)	At the end of the production line –after completing topping- just before the packaging machine. Alarm to activate manual sorting of suspicious unit for further detailed visual inspection.
PLATOS TRADICIONALES	Ready-to-Eat (RTE) foods such as lasagne, rice dishes and roast dishes	Plastic pieces and films, rubber, thread, insects, stones, wood, paper, hair, etc.	At the end of the production line just before the packaging machine (sealing). Alarm to activate manual sorting of suspicious unit for further detailed visual inspection.
MARÍN GIMENEZ	Processed fruit and vegetables	Fragments of fruit pits and kernels, worms and larvae	Once the fruits have been cut in dices, just before packaging the product in cans. Identification of the suspected body for removing it and the surrounding bodies by means an existing sorter.
DAWN FARM FOODS	Cooked and fermented meat ingredients	Plastic pieces and films, rubber and fragments of bones	At the end of the production line just before the packaging machine (sealing). Alarm to activate manual sorting of suspicious unit for further detailed visual inspection

Using a feasibility prototype system that is available in IRIS (see figure 2 above) we will undertake an initial demonstration of the system on the applications to provide test data for the development project. An iterative product development cycle will be implemented to design and produce beta prototype systems that will be installed at each of our end-user sites as test cases for performance verification, the success of which will confirm the acceptance of the prototype designs. Having validated these designs we will

translate the prototypes to product through the creation of the manufacturing documentation package using the engineering information and specifications and we will set up our manufacturing facilities and produce and quality approve the first units. In parallel to the product development activities we will also implement **market development activities,** whereby based on planning within the framework of the classical 4 Ps (Product, Pricing, Promotion and Place) Marketing Mix, we will perform comprehensive activities to gear up for the Commercialisation phase.

In relation to methodologies that will be used during the development work, our teams operate under agile frameworks, whereby Kanban is employed for hardware developments, and Scrum and Test Driven Development (TDD) for software developments. Whenever creative thinking is needed, particularly for ideating specific solutions that cannot be addressed by finding a simple tradeoff among several technical constraints, the development will be based on TRIZ (Theory of the resolution of invention-related tasks), which is a widely recognized methodology for accelerating and simplifying the search for inventive and innovative solutions and a scientifically founded approach to the forecasting of the evolution of technological systems, products and processes[11]. The IRIS TRIZ Team has been properly trained by recognized members of ETRIA (European TRIZ Association[12]) for undertaking creative challenges associated to R&D tasks. The TRIZ approach will be particularly convenient for finding flexible (versatile) designs compatible with the most of the applications to be addressed by HYPERA.

HOW THE PROJECT CONCEPT AND OBJECTIVES FIT INTO THE OVERALL PLAN TO REACH THE MARKET

The HYPERA concept and the planned development is oriented towards market takeup; so this is a core business objective and the ultimate goal of the project. Our application-focussed approach will enable us to gear up for sales in foreign body detection in food processing applications which will be supported by clearly presented application notes that clearly demonstrate the technical performance and benefits of the use of the HYPERA system. These validated applications will demonstrate: reliability, repeatability, the robustness of the device, its operational and industrial-level performance, its translation to product and establishing the manufacturing are all important steps of a development on its way towards commercialisation. With an injection of funds, and by gearing up right from system specifications, through design, prototyping, industry trials in compliance with all industry standards and regulatory requirements, we will ensure that there are no user-, technical-, market- or regulatory- obstacles to the uptake of the technology in the marketplace.

Finally, our commercialisation-related activities will also be fundamental towards leading to effective market takeup. As such, IRIS will set up an Applications Team to support the Sales Team (this is important given that we are dealing with a process analytical system that will always require some level of customisation for a specific application, as well as specialist installation and ongoing support). We will also validate our pricing strategy, prepare our sales agent/distribution networks (our existing distributorships and new ones), and prepare all aspects of our market launch (first devices, replacement stocks, catalogues, datasheets, pricelists, etc.).

HOW THE PROJECT DEVELOPS SOMETHING NEW TO EUROPE AND ADDRESSES EU-WIDE CHALLENGES

Food safety concerns, accentuated by recent crises, have undermined public confidence in the capacity of the food industry and public authorities to ensure that food is safe, and are therefore **high on consumer, political and food industry agendas worldwide**[13]. Assuring food safety and improving quality in the European food supply chain is of primary importance. The EC has identified food safety as one of **its top priorities** and, with Regulation (EC) No 178/2002 adopted by the European Parliament and the Council on 28 January 2002, it laid down the general principles and requirements of food law. **Food security** is also identified as one of the key societal challenges to be addressed during Horizon 2020[14]. The ability to leverage the SME Instrument to accelerate the transformation of new knowledge for industry to conceive and develop new products, industrial processes and services, is an important tool for creating a true Innovation Union: a European Union that thanks to its research and innovation performance can support sustainable growth. The HYPERA® project will output a European developed and manufactured specialised food assessment hyperspectral imaging system that will respond to growing demand in Europe and globally for automatic food inspection technology that is cost effective, reliable, and is capable of keeping up with the speed of modern food processing lines.

From the **technology side**, photonics is regarded as a driver of innovation and plays an important "enabler" role for many industrial and customer markets. With a worldwide market value projected to hit €615 billion in 2020 and to remain stronger than the gross domestic product (GDP), photonics has moved from a niche application to a key enabling technology, according to Photonics Industry Report 2013. The European photonics industry holds about 55 % of the world market in production technology. HYPERA® will advance the suitability of HSI for food processing applications in spectroscopy to increase EU competitiveness in the photonics field by, as was seen in Table 1 earlier. By scaling up and integrating these novel combination of technologies into the HYPERA® system, we are bringing a technological evolution in photonics to lower the cost of the technology so that HSI spectroscopic technology can be broadly adopted as a system for the real-time monitoring of quality, safety, as well as the nutritional (through chemical composition analysis)

[11] TRIZ. Innovation and Inventive Problem Solving Handbook", Pavel Livotov and Vladimir Petrov ©TriS Europe, Berlin 2014. www.tris-europe.com
[12] http://etria.eu/portal/
[13] http://ec.europa.eu/research/bioeconomy/food/policy/index_en.htm
[14] Brussels, 30.11.2011. COM(2011) 808 final.. Horizon 2020 - The Framework Programme for Research and Innovation

and commercial properties of a wide variety of food products during all steps along the production chain. As such, the innovation impacts of this project are high, as the knowledge generated will be industrialised and taken to market by IRIS.

From the **application side**, a technology that can increase food quality and safety, protect food producers and their brands, and ultimately contribute to raising consumer confidence in the food industry is of high added value for Europe, especially when we consider the importance of the food supply chain in the context of the European economy, whereby it connects the agricultural, food processing and distribution sectors that together constitute more than 7% of European employment. More specifically, the EU food and drink industry alone provides jobs for more than 4 million people in 310,000 companies, generating a turnover in excess of €965 billion each year and serving over 500 million consumers[15]. However, growth in value-added and labour productivity within the EU food industry is **outpaced by that of the United States** and at world level, EU food products are **losing market share to Brazil, Canada and Australia**. HYPERA® will contribute to **leadership in enabling and industrial technologies**, which is synergistic with one of the priorities of H2020 for contributing to **industrial leadership** in Europe. To survive in an increasingly globalised and competitive market, agro-related enterprises have to form and count on close associations with clients and suppliers, whereby product quality and safety are increasingly intrinsic with such relationships. In fact, the EC appointed High Level Group on the Competitiveness of the Agro-Food Industry[16] has urged the EU to maintain a **high level of food safety and quality standards**, and has identified a strong need for an improved innovation framework and targeted actions on the numerous SMEs in the sector and on the performance of the EU agro-food industry on world markets. Solutions such as HYPERA®, which are founded on a growing need faced by industry and the potential to apply knowledge for obtaining ambitious technological advances, to bring to market technologies that are accessible to wide communities of SMEs and can and improve the competitive capacities of the EU food and drink industry, will play a central role in treading this line between guaranteeing consumers the safest and most affordable food products possible while also ensuring that Europe's food sector is as efficient, competitive and balanced as it can be. As was mentioned, technologies such as HYPERA will help companies transition to the Industry 4.0 paradigm.

HOW GENDER ANALYSIS IS TAKEN INTO ACCOUNT IN THE PROJECT'S CONTENT

We will consider gender related design features, whereby the needs of female line operators will be analysed in cooperation with our end-user early adopter companies, so that any gender-specific design features can be embedded. As such, during the co-creation process in WP1, female engineers/developers, as well as female professionals from the food industry will be involved to ensure gender related aspects are picked up. Moreover in relation to business behaviour and decision making, gender analysis will also be important, especially in view of growing trends and commitments to see more women occupy high level and decision-making positions in industry. There is also a gender consideration in relation to our communication activities towards consumers, whereby our ability to successfully inform and generate awareness among female consumers (many of whom are still key household consumption decision makers) will be important. It is noteworthy that our Project Manager is a highly qualified female professional (Dr. Edurne Gaston), who is also a recent mother. Numerous other members of the team are female, such as the Commercial Director (Johanna Granda), among key researchers and engineers who will be working on the project (see section 4.1). With regard to the employment opportunities that will be generated in IRIS for the implementation of the innovation project, and subsequent jobs through the commercialisation of the project outputs, a recruitment policy will be in place designed to ensure a reasonable level of balance between the sexes.

1.4 AMBITION

NOVELTY OF THE INNOVATION BUSINESS PROJECT

In terms of a 'business project', IRIS has the ambition to bridge the **'integration gap'** that prevents the adoption of photonics in industry. The typical HSI landscape in Europe is made up of very competent providers of imaging cameras (such as INNOSPEC, SPECIM), providers of chemometric software (such as Perception Park), along with distributors of cameras and chemometric software. There is a distinct lack of solutions providers **offering turnkey solutions to industry**. IRIS has identified a clear market opportunity, not only among the SME sector but also among the enterprises sector as well, for customised monitoring technology that shifts the emphasis from "device" to "fully integrated turnkey solution" based on tailoring the device to the line and to the specific operating environment of the client. Key to mainstreaming affordable monitoring solutions for industry is building **flexible and reliable platforms based on novel combinations of spectroscopic analysers and chemometric data processing tools** that can effectively analyse the data, to deliver fast and reliable readings in real-time, to facilitate rapid calibration of the HYPERA® hardware that lends itself to easy set-up and customisation to specific client inspection and monitoring needs and industrial environments.

Our vision is for the HYPERA® brand to become a leading global reference for **highly affordable, rapid, accurate and high performance real-time food quality inspection turnkey solutions** that are supported by reliable integration, data management and maintenance services. Such a vision is fundamental for the widespread uptake of HSI equipment in the industry, whereby food processors typically lack the specialised in-house resources to be able to confidently invest in and set up high-end spectroscopy real-

[15] Nurturing the competitiveness of the European food industry (07.10.10) Enterprise & Industry magazine (http://ec.europa.eu/enterprise/e_i/index_en.htm), © EU, 2008 - 2010
[16] Commission Decision No 2008/359/EC of 28 April 2008.

time inspection systems. HYPERA® will position IRIS in the global food industry as a leading **European solutions provider** of affordable, fully integrated HSI systems for the rapid, non-destructive detection of foreign bodies and chemical composition analysis in foodstuffs.

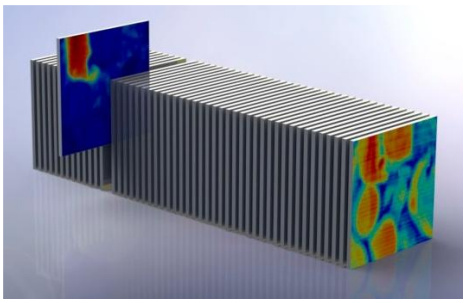


FIGURE 3- HYPERCUBE

EXPECTED MARKET APPLICATION THAT DIFFERENTIATES THE PROJECT AND PROVIDES THE HIGHEST ADDED VALUE

As has been discussed, our application oriented development is centred on food inspection. The **original and innovative nature of the HYPERA® system** is hinged on the necessity to provide the food industry with **an affordable, high performance, rapid, accurate and multifunctional HSI inspection system that can be easily integrated into food production lines** (and in quick succession the pharmaceutical industry) for the detection of foreign bodies and for chemical composition monitoring during food processing. The system is founded on the promising results to TRL6 of the highly novel combination of Shortwave Infrared Hyperspectral Imaging (SWIR-HSI) in reflectance with Near Infrared Multispectral Imaging (NIR-MSI) in reflectance/transmittance and integrated chemometric software.

Recently, optical sensing technologies such as spectroscopy and imaging have been investigated as potential tools for the non-destructive evaluation and inspection for food quality and safety. Spectroscopy is a widely recognised technique for quality inspection based on the measurement of optical properties of food products (Cen & He, 2007). Near Infrared Spectroscopy (NIRS) is **rapid, non-destructive, and relatively easy to implement for on-line and off-line applications**. More importantly, NIRS has the potential for **simultaneously measuring multiple quality attributes**. However, while with these spectroscopic techniques it is possible to obtain information about the sample components based on the light absorption of the sample, it is **not easy to know the position/location information**. On the other hand, measurement of the spatial features of food products can be achieved by conventional imaging system or more specifically computer vision with the minimum of human intervention (Du & Sun, 2006b). Yet, with **computer vision systems**, while it is **easy to know the position of certain features**, it is not easy to conduct the quantitative analysis of a component. Moreover, because it is operating at visible wavelengths in the forms of monochromatic or colour images, a conventional imaging system is incapable of inspecting specimens with similar colour, classifying complex objectives, predicting internal attributes (e.g. chemical components) that are less relevant to colour, and in detecting defects not visible to the human eye.

As seen, **Hyperspectral imaging (HSI)** combines imaging and spectroscopy for enhanced non-destructive analysis of materials[3]. Looking deeper at this concept, for a hyperspectral image, the complete wavelength spectrum of a sample over the spectral range is obtained at each point in the imaging plane. On the basis of the combination of the series of images, a fourth dimensional "hyperspectral cube" can be established (figure 3), which includes one dimension representing spectral information, two other dimensions for describing spatial information and the last dimension contains the optical signal level (sample radiance). In this cube, each 2D-image point (or pixel) has both spatial and spectral signature (spectrum) information. Therefore, HSI has the capability **to capture both physical and morphological characteristics such as colour, size, shape, and texture, and some intrinsic chemical and molecular information (such as water, fat, and protein) from a food product**. When combined with chemometric (i.e. multivariate) analysis, HSI can be applied for prediction of the concentration and distribution of multiple components in a sample[3]. This makes the technique very attractive for foreign body detection in food processes, whereby HSI can detect, as well as classify, foreign bodies according to their spectral properties. Recently, the technique has been widely studied for its applications in the food industry, from non-destructive assessment of mushroom quality and shelf-life prediction[17], objective and quantitative determination of meat quality, including beef, pork, and lamb[18], inspection of moisture and fat contents[19], nematodes[20], and blood spots[21], freshness detection[22] and production line sampling[23] in fish (to mention a few).

Indeed, since its development over 20 years ago, HSI has gradually expanded from its initial use as a research tool for government and mining industry use into more **commercial application areas found in the industrial, agricultural, geological, environmental and medical communities**. However, this transition has been slow, because most of the research has focused on addressing the high-performance demands of federally-designated tasks, such as airborne and space-based military surveillance. As a result, until recently, there had been little real investigation of the commercial marketplace because manufacturers of optical equipment were focusing on

[17] A. A. Gowen, C. P. O'Donnell, M. Taghizadeh, E. Gaston, A. O'Gorman, P. J. Cullen, J. M. Frias, C. Esquerre and G. Downey. Hyperspectral imaging for the investigation of quality deterioration in sliced mushrooms (Agaricus bisporus) during storage Sensing and Instrumentation for Food Quality and Safety Volume 2, Number 3 (2008), 133-143, DOI: 10.1007/s11694-008-9042-4

[18] D. F. Barbin, G. Elmasry, Da-Wen Sun, P. Allen, Predicting Quality And Sensory Attributes of Pork Using NIR Hyperspectral Imaging, Analytica Chimica Acta, 719 (2012) 30– 42.
G. ElMasry, Da-Wen Sun and Paul Allen, Near-Infrared Hyperspectral Imaging for Predicting Colour, pH and Tenderness of Fresh Beef, J.l of Food Engineering, 110 [1] (2012) 127-140. M. Kamruzzaman, G. ElMasry, Da-Wen Sun and Paul Allen, Prediction of Some Quality Attributes of Lamb Meat Using NIR Hyperspectral Imaging and Multivariate Analysis, Analytica Chimica Acta, 714 (2012) 57-67.

[19] ElMasry, G. & Wold, J.P.(2008) High-speed assessment of fat and water content distribution in fish fillets using online imaging spectroscopy. Journal of Agricultural and Food Chemistry, 56(17), 7672-7677

[20] Sivertsen, A.H. (2011) Automatic nematode detection in cod fillets (*Gadus Morhua*) by transillumination hyperspectral imaging. Journal of Food Science, 76(1), S77-S83

[21] Heia, K., Sivertsen, A. & Birkeland, S. (2009). Automated quality control of salmon fillets - hyperspectral imaging for on-line inspection. In (Eds.) *14th international conference on near infrared spectroscopy*

[22] Menesatti, P., Urbani, G., Pallottino, F., D'Andrea, S., & Costa, C. (2007). Nondestructive multi-parametric instruments for fish freshness estimation. Instrumentation Viewpoint, 6.

[23] Wold, J. P. (2006). Non-contact transflectance near infrared imaging for representative on-line sampling of dried salted coalfish (bacalao). Journal of Near Infrared Spectroscopy, 14(1), 59–66.

the technical challenges of the federally-driven application segment, which had still not been fully addressed[24]. To date HSI technology has only **been directly implemented to a very limited extent in on-line systems for automated quality evaluation purposes.**[25]

Of the few food quality control applications, these have been directly developed by the generic HSI camera manufacturers –mainly for demonstrating the possibilities of the systems, that is, for opening market opportunities- as opposed to by system integrators oriented to satisfy specific industry needs. As was seen key limitations are centred on the speed of the imaging acquisition, which needs to be increased to keep pace with the rapid rate of novel food processing lines. Moreover, current HSI systems are expensive due to the bespoke nature of existing solutions as well as the predominance of research and high value targets in the current HSI market, with a typical system integrated in a production line costing up to €150k; prohibitive to the majority of SMEs from the food sector, not to mention enterprises running numerous lines. As was discussed earlier in the document and summarised in Table 1, we will bring a number of key advancements that will open up the food inspection application to HSI technology. To recap briefly on these, not only will we use very cost-effective imaging cameras (especially as the resolution requirements are not as high as typical applications, such as forensics and military applications, where HSI has been largely applied) that are specifically designed for application in food production lines. As was seen, hyperspectral images contain a substantial amount of data, including much superfluous information, **posing considerable computational challenges**[25] that involve lengthy image acquisition and analysis times, rendering current HSI systems as impractical for implementation on-line in modern food processing facilities. To solve this problem, the HYPERA® system will be designed based on the understanding that we do not need the image with the whole area and the whole spectrum to assess the food quality (on the contrary, only a small number of spectral bands and a special region of interest (ROI) will be enough), thereby enabling the ability to **increase the hardware speeds** for rapid image acquisition and analysis to ensure compatibility with typical processing speeds of **1 m/s**. The spatial and spectral resolution requirements can also be reduced compared with general hyperspectral imagers.

Moreover, as was discussed, HYPERA® combines NIR and SWIR ranges in two solidary but independent modules that sum up the advantages of each region as each module is optimized in terms of lighting, signal gain and raw data processing methods, and the extended detecting spectral range will give the chemometric tools more available spectral bands to choose from and will allow the system to integrate a variety of chemometric methods and select different ones for various purposes and objects, thereby realising multifunctional inspection. These combined observing bands and the processing methods have, to date, **never been used in food (or pharmaceutical) production applications**. What is currently found in the market are simply just extended cameras (VIS+NIR+SWIR). They are tuned to reach a trade-off among such ranges. Consequently, they do not take advantage of each range-specific features: so if is sensitive in SWIR, the NIR signal is low; if sensitive in NIR, it is saturated in the SWIR…It is pretty much like putting wings on a car: too slow for flying and too wide for any street.

In short, this project will design a hyperspectral imager that simply meets the requirements with no (or few) surplus performances, enabling significant cost reductions, as well as greater speed and flexibility. The resulting HYPERA® system is on the cutting-edge of existing efforts in the field to produce reliable tools for food/pharmaceutical quality and safety assessment. Its unique advantages provide a versatile and highly cost-efficient solution to the problem of applying HSI technology to the real industrial world.

EXPECTED PERFORMANCE/IMPACT ON DEFINED NEEDS

The ability of HSI to provide **rapid and objective evaluation of various product quality and safety indicators simultaneously** is highly valuable for the food industry, and greatly advances the capability of computer vision-based technologies, which are **inefficient in differentiating samples that present similar colour characteristics but are not of the same quality**. Since such vision systems are operated at visible wavelengths, they can only produce an image registering the external view of the objects, and are unable to register information related to the **internal structure or the chemical composition of the products**. HSI opens up the opportunity to register product features that are sensitive to wavebands other than visible, and use this information for **the non-invasive and non-destructive assessment of quality and safety of foods**.

HYPERA® will be the first HSI complete solution that is specifically designed for affordable and seamless integration into food processing lines for foreign body detection and chemical analysis of foodstuffs (in addition to pharmaceuticals processing and other applications in the future). Major advances that will be offered by our HSI food inspection and foreign body detection system vs. competing HSI systems (discussed in Section 2) are the **reliability, simplicity and affordability of the proposed HYPERA® system**. Compared with current quality assessment hyperspectral imagers, there are 5 essential innovations of the proposed system (Table 5) which will fulfil industry requirements to ensure the HYPERA system is suitable for use as foreign body detection system in modern food processing.

TABLE 5- UNIQUE FEATURES DELIVERED BY HYPERA®

Desired feature	Why is this important?	How the HYPERA solution ensures this feature

[24] Bannon, D. and Thomas, R. Meeting the Optical Demands of Next Generation Hyperspectral Imaging Spectrometers. Source: http://www.headwallphotonics.com/Portals/145999/docs/photonics_tech_briefs.pdf
[25] ElMasry, G., & Sun, D.-W. (2010). Principles of hyperspectral imaging technology. In D.-W. Sun (Ed.), Hyperspectral Imaging for Food Quality Analysis and Control (pp. 3-43). London, Burlington, San Diego: Academic Press, Elsevier. ISBN: 9780123747532

Penetration	Many foreign bodies are buried in the food material and are not simply present on the surface	NIR-MSI enables penetration of food matrixes to a typical depth of 5 cm- The use of chemometrics –even in such poorly informative spectral regions- contribute to address in some extent the issues caused by scattering: while shape and position information is blurred by scattering, the scattered rays hold composition-related information that can be unveiled by powerful chemometrics methods-**this is a major advance for foreign body detection.**
Composition differentiation	There is a need to be able to detect and classify foreign bodies, their type, vs. food materials that should be present in the food; contrast is better in terms of composition rather than in terms of color.	SWIR spectral ranges allow for better composition differentiation enabled by the ability of the extended infrared spectral bands to obtain more chemical information from the sample.
Accuracy	Image distortion caused by scattering is one of the key challenges with regard to the detection of bulk-borne bodies; even when transparency is not a problem many substances show poor optical absorbance in the NIR and SWIR ranges- the irregular nature of many bodies degrades the spatial information and the shape and position become undefined.	Chemometric models will minimise the consequences of the image distortion caused by scattering, as explained above. For artificial vision and X-ray systems, defined shape is a key requirement because the detection is just based on identifying unexpected shapes (shades, refractions and reflections, for example). Chemometrics works with spectral data instead. Moreover, redundant (corroboratory) information –provided by two different but complementary spectral ranges, NIR and SWIR- increases detection (identification) probability and reduces uncertainty.
Compactness, low-cost design	Low cost tolerance in food industry for equipment, installation and operational investments.	Enabled by the extended infrared spectral bands to obtain more chemical information from the sample using novel detectors.
High inspection speeds, affordability and versatility	Need to be able to keep pace with modern processing lines. For retrofitting purposes- seamless installation into existing lines causing as little disruption as possible, as well as easy access for maintenance. Software upgrades should enable additional products and objects to be inspected.	By combining a flexible hyperspectral imager design with integrated chemometric tools based on the in-depth analysis of industrial requirements to adapt multiple application purposes and various objects, tackling one of the major reasons for the high cost and slower speeds of current HSI systems, whereby current food inspection hyperspectral imagers typically acquire much superfluous information which occupies a large part of the system resources.

While currently available NIR and SWIR HSI systems offer good results in the laboratory, the HYPERA® system, with extended infrared spectral range, will **offer improved qualities, lower cost and new features** to be welcomed and widely used by the industry production lines. The chemometric tools combined with the hyperspectral imager proposed in the present application will address the long standing problem of **low efficiency and single function**.

When compared with other competing inspection systems in use in the market today, such as metal detection and x-ray systems, in relation to foreign body detection, each product presents special technological challenges. For example, translucent and opaque liquids filled in bottles, coloured opaque plastic packages, tin cans or milk cartons all respond differently to available detection systems. Metal detectors find splinters from machinery, fractions of broken cutters and blades, needles, screws or fragments of clips. X-ray detectors respond to metal, stone, bone, hard plastics and Teflon. Both systems can screen the product after the filling procedure. X-ray detectors may find glass pieces that result when a jar or bottle is crushed during malfunction of the packaging line. More recent developments such as near-field radar response enables foreign bodies to be detected by transmitting low-power microwaves through the material[26], whereby the dielectric properties of the food flow of processing equipment is measured. Foreign materials produce a deviation from the norm, activating a pneumatic rejection unit. However, as was discussed earlier, some of these methods are expensive and not versatile enough for offering comprehensive quality control. This opens up enormous opportunities for our HYPERA® solution, particularly when the unwanted bodies are very similar –in terms of composition and density- to the foodstuff. Moreover, HSI is more sensitive than the homologue "1-pixel" spectrophotometer because of a geometric fact: in the event of a small body whose chemical composition differs from the composition of the surrounding bodies, its spectrum will be "pure" and, consequently, easily identified; on the contrary, by averaging the whole image in only one optical channel –as the typical "1-pixel" spectrophotometer does-, the contribution of such small body is not relevant and the body will not be detected.

Another key advantage of the HYPERA system over competing HSI systems, its ability to penetrate materials. In terms of transparency, can be partially addressed in the SWIR range –around 5 mm or deeper according to each particular case- and even deeper in the NIR range -5 cm or more depending on the case. VIS imaging (artificial vision) is not capable of penetrating more than 1 mm. of materials.

Finally, in terms of **improvement potential over time**, the ability to completely eliminate the image distortion that is created due to light scattering would heighten further the competitive advantages of the HYPERA system. As foodstuffs are not homogeneous media in the widest sense, such non-homogeneities causes scattering and, consequently, loss of spatial information (position and shape). While our chemometric tools will significantly reduce this scattering problem, our ambition is to completely eliminate this over time, through the use of highly novel Imaging in Complex Media (ICM) tools, which will perform special mathematical pre-processing for overcoming this drawback. Previous work with ICM on the analysis of seismic waves and surveillance through foggy media, improved by means of the newest "deep learning" solutions for extracting trends (order) from chaotic sources of data (big data paradigm) will be built upon in order to continue to add competitive advantages and breakthroughs to strengthen the market competitiveness of the HYPERA product.

[26] Swedish Institute for Food and Biotechnology and the Food Radar System (FRS) in http://www.foodradar.com/assets/pdf/fruit_processing_0112.pdf

2. IMPACT

2.1 EXPECTED IMPACTS

Foreign body hazards represent one of the most significant risks associated with the production and distribution of food products. Prevention is key to minimising the risks to consumer health and safety and avoiding product rejections and recalls that can lead to high levels of food being wasted. The HYPERA® system for the cost-effective, rapid, reliable and non-destructive real-time inspection of foods for foreign body detection and chemical composition analysis. This non-invasive quality control method for preventing non-spec products from entering the food chain, will avoid product rejections, product recalls, and will safeguard consumer health and safety. It is a timely development that will deliver high impact.

A) USERS / MARKET

USER NEEDS THAT ARE BEING MET

This project responds to a real and growing **need among food processors, both large corporations and SMEs to guarantee the quality and safety of their food products and to prevent foods containing foreign bodies from entering the food chain**. Indeed, the production of safe food is considered one of the most challenging jobs in the world[27]. All food producers, distributors and retailers are surrounded by risk. Foreign body contamination of food and drink products is the single biggest cause of product recalls[28]. Brand protection is an increasing preoccupation with food processors and the wider food supply chain, whereby central to the ability to protect a brand, is the ability to avoid a recall in the first instance. In modern food processing, the number of occurrences of foreign materials is decreasing as the food industry continuously improves the safety and quality of their products. However, this trend also decreases public acceptance for such occurrences to a level that approaches "**zero tolerance**."[29]. There is increasing pressure on the food industry to adopt self-regulation for food quality and safety best practices and standards and are becoming critical to **food processor's business success**. Any incidence of foreign material harms the consumer, undermines confidence in the brand and generates headlines. The name of the producer and the grocery chain that sold the article are widely reported.[29] The ability to manufacture high-quality products consistently is the basis for success in the highly competitive food industry. It secures brand protection and encourages loyalty in customers and the ability to attract new customers to increase market share. Applying a rapid, accurate, and non-invasive technique in food analytical procedure for quality evaluation and safety inspection has become an urgent need in the food industry[10]. Beyond the European context, globally food producers are also facing mounting pressure. In the US for example, the food supply need to take actions in line with the Food Safety Modernization Act (FSMA) that mandates all processing facilities implement comprehensive, science based preventive controls that are appropriate for their operations. With increased consumer demand, and market and legislative expectations for food products with high quality and safety[30], the need for accurate, fast, and objective quality determination of these characteristics continues to grow. Moreover, in modern food processing, as plant throughput increases and quality tolerance is tightened, it is necessary to employ **automatic methods for quality assurance and quality control**[31].

Four large and SME food processors- CASA TARRADELLAS (pizza processor), PLATOS TRADICIONALES (Ready to Eat foods), MARÍN GIMENEZ (processed fruit and vegetables); DAWN FARM FOODS (cooked and fermented meats)- will collaborate with us in an early adopter capacity as they all, despite having different systems installed, such as X-ray systems, are seeking complementary and robust solutions in order to overcome the limitations of their existing systems and to reduce the burden on visual inspection.

[27] http://www.foodindustrycounsel.com/2015/05/protecting-your-brand-by-finding-and-eliminating-food-safety-risk/
[28] Ray Gibson, (2000) "Foreign bodies cause most recalls", Nutrition & Food Science, Vol. 30 Iss: 4, pp.187 - 191
[29] http://www.foodsafetymagazine.com/magazine-archive1/octobernovember-2012/testing-foreign-object-detection-integration-in-food-production/
[30] Nagata, M., Tallada, J. G., Kobayashi, T., & Toyoda, H. (2005). NIR hyperspectral imaging for measurement of internal quality in strawberries. Paper No. 053131, ASAE Annual Intl. Meeting, Tampa, Florida, US.
[31] Gunasekaran, S. (1996). Computer vision technology for food quality assurance. Trends in Food Science & Technology, 7(8), 245–256.

An assessment of industry needs has been central to the conceptualisation of the HYPERA HSI innovations (see Table 1) and in the unique features that the product will offer (see Table 5), whereby the key requirements that need to be met, according to our early adopter companies and a wider consensus with companies that have been interviewed from our client base and networks, as well as industry engagement at following trade exhibitions and conferences: BTA Barcelona Food Technologies (May 21-24 2015), Global Food Safety 2015 in Amsterdam (September 8-9, 2015), Anuga Food Congress in Cologne (October 10-14 2015) are as follows: **a) speed** (to keep with processing speeds of 1m/sec); **b) affordability** (in terms of equipment cost, installation and operating costs- € 80k for an installed unit will give a ROI in 2-3 years in terms of saved QC costs); **c) accuracy** (>95% in classification and better than 10% around the target value are considered acceptable accuracies for online applications).

With the continued technical innovations in manufacturing and computing, it is now possible for us to develop a HIS system that is capable of performing in a low-cost and high-speed way for online and real-time detection of various products[10].

MAIN ECONOMIC BENEFITS FOR THE USERS AND UNIQUE SELLING POINTS

To be profitable, industry requires equipment that ensures high-quality production, and efficient work methods and cost. The ability to automate the quality control and inspection process will save time compared to manual or post-production product inspection, thereby increasing product throughput, involving less handling (and subsequent reduced risk of food infection with microbes and pathogens), reduced incidences of spoilage as it is a non-invasive method, as well as **removing human subjectivity and inconsistency**. This fact is even more important in developed countries to enable companies to defend their position and competitiveness against others where labour cost does not account for such a significant part of the overall manufacturing cost[32]. Considering that the **productivity of a person working in a mechanised and automated environment is approximately ten times that of a manual worker**[33], if quality evaluation is achieved automatically, production speed and efficiency can be improved drastically in addition to increased evaluation accuracy, with an accompanying reduction in production costs. Moreover, automation usually increases the productivity and changes the character of the work, making it less arduous and more attractive.

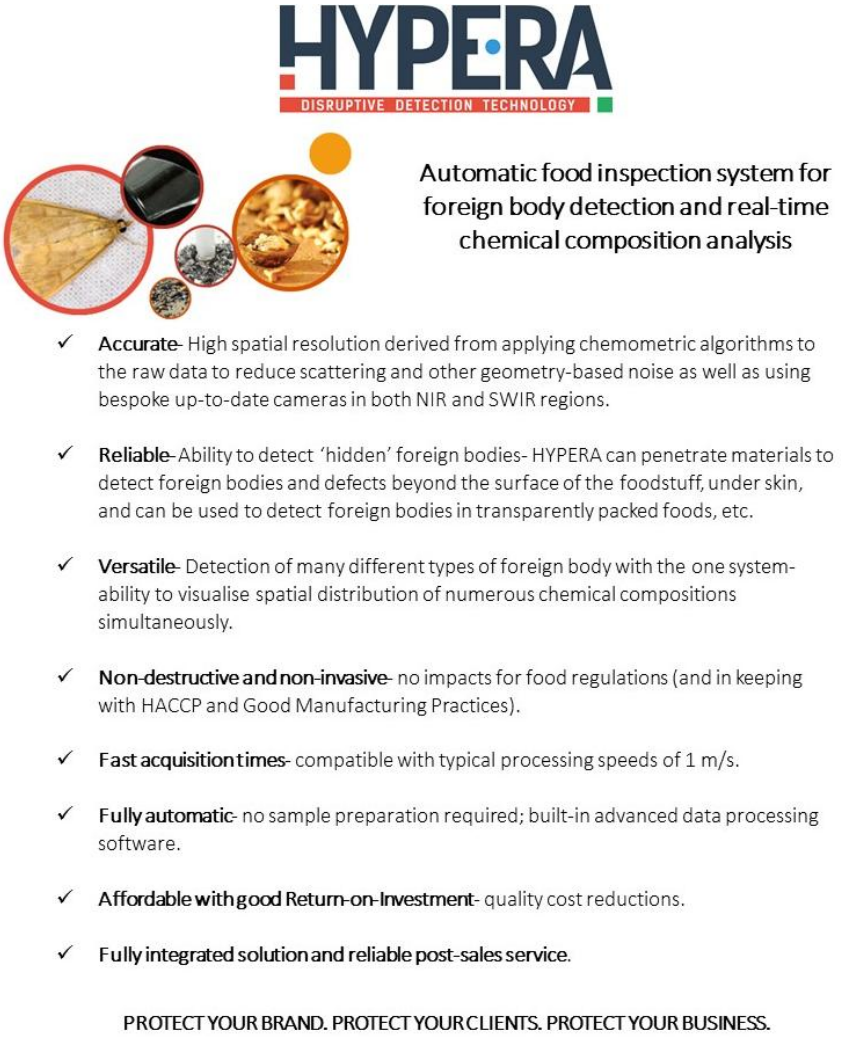


FIGURE 3- POSTER OF THE HYPERA USPS.

If we consider, the costs of a product recall, HYPERA® system is a very worthwhile insurance policy: recall costs, which include getting food off shelves, handling lawsuits, repairing public relations, and potentially revamping plants can pose an enormous burden on companies, in addition to damaged reputation and lost sales. The potential negative impact of publicity surrounding an incident means that manufacturers have to be meticulous, and their systems need to provide confidence that any complaints cannot have originated in their factories. If there is any doubt at all then this can lead to costly and damaging product recalls. For smaller companies, the ensuing economic impacts can result in them going out of business.

The **Unique Selling Points** of the HYPERA® system that will be promoted to Quality and Production Managers, and other key decision makers in the food industry, are summarised in the poster in Figure 3. These USPs (which will be revised and upgraded as necessary in the early stages of our Phase 2 project, and used as a communication tool towards the industry) will contribute to strong

[32] Applications and Experiences of Quality Control. Chapter 33: Hyperspectral Imaging for Raw Material Sorting and Processed Product Quality Control. Pilar-Beatriz García-Allende, Olga M. Conde and José Miguel López-Higuera. University of Cantabria Spain.
[33] Abdullah, M.Z.,Guan, L.C., Lim, K.C., & Karim, A.A. (2004). The applications of computer vision system and tomographic radar imaging for assessing physical properties of food. J. Food Engineering, 61, 125–135.

technology push as major drivers for stimulating market demand for the technology among food processors, opening up an enormous opportunity for HYPERA® to secure a leading position as the most complete, high performance HSI equipment in the market today and will make the system highly suitable for easy integration and use in industrial food processing lines for the real-time detection of foreign bodies and chemical analysis of foodstuffs (and other products).

DESCRIPTION OF THE MARKET

We are addressing the **global food processing market**. Looking at the size of the market, in the context of the Europe alone, the food and drink (F&D) industry is the largest manufacturing sector both in terms of turnover (€1017 billion), Value Added (€203 billion) and employment (generating 4.25 million jobs)[34]. SMEs make up 285,000 (or 99.1%) of the total food companies in the EU[34]. Globally the industry is increasingly concerned with food safety, as highly publicised, high profile food safety incidences and scandals in recent years have undermined consumer confidence and trust, and more stringent legislation and market pressures are forcing the industry to improve quality control and safety standards and to take measures to protect their brands.

In **Europe** the following scandals (to mention a few) are defining a market whereby food quality and safety are central concerns: the recent horse meat scandal in beef products that has disrupted the food chain in a major way throughout Europe; the 2011 E.Coli outbreak where some 30 people died and over 3000 people were sickened; the mad cow disease that wreaked havoc on the European beef industry in the 90's (whereby the effects are still present and anybody that spend at least 3 months in the UK between Jan 1 1980 and Dec 31st 1996 is prohibited from donating blood). In the **global context**, in the People's Republic of China for example, food safety is of growing concern, fuelled by the high publicised cases of deaths and illnesses caused by melamine-laced milk and baby formula (in 2008 when at least six children died after consuming milk that contained melamine, an industrial chemical that causes kidney stones in children), Salmonella-tainted seafood, and clenbuterol-treated pork. The misuse and overuse of hundreds of chemicals, many of them banned and toxic, have led to death for scores and sickness in hundreds of thousands of people throughout China. Recent opinion polls reveal that a majority of ordinary citizens mistrust their nation's food suppliers and processors[35]. The situation has become a matter of grave concern. The problem affects not only China but also Europe, the US and other countries that import Chinese-grown and processed food. In **the US**, in 2011, the listeria outbreak in cantaloupe was noted as the deadliest outbreak of food-related illness in more than 10 years in the United States. 30 people died and 146 were sickened. In 2012, there was another big cantaloupe recall from a different farm. The "pink slime" scandal, a controversial beef additive made up of spare beef trimmings that have been treated with ammonium hydroxide to make them safe and at least semi-palatable shows the power of social media and how quickly news can go viral and brands can be irreparably damaged. Celebrity chef Jamie Oliver brought the "pink slime" term to light, and while the additive may have been safe enough for consumption, fear-mongering coupled with distrust of the beef industry led to a massive outcry, and the closure of three plants.

Indeed, the food supply chain has become a complex global system consisting of small to large domestic and foreign manufacturers, processors, packers, distributors, and transporters with few common business practices[36]. As retailers, food service organizations, and marketers of branded products invest to promote their brands in the marketplace, many food processors are now being measured on their ability to help customers protect their brands, reputations, and profitability[37]. Consequently, in addition to traditional metrics such as price, product consistency and service, **brand protection is increasingly being used as a new competitive metric to earn and keep customer's business and loyalty**. **Food processor who can demonstrate the most reliable means of brand protection can leverage a significant competitive advantage in the value chain**. Food processors who can demonstrate they are taking measures to ensure food safety and quality are in a position to command favourable pricing and enjoy "preferred supplier" status with customers, while also benefiting by raising the barrier to entry by competitors and protecting their own existing market share[37]. Real-time quality monitoring technology, such as HYPERA® can play a key role in brand protection by preventing off-spec products entering the supply chain. These are all major drivers for investment in the HYPERA® solution. Importantly, investments in equipment are increasing: **Investment in food processing machinery has been climbing ~7.3 % per year since 2011-2012 and is predicted to reach $53.3 billion in the global market in 2016[38].**

MAIN COMPETITORS AND THEIR COMPETITIVE SOLUTIONS

A number of manufacturers of HSI equipment exist in the market today and Table 6 provides a comparative overview. Two of the HSI equipment manufacturers from Europe- SPECIM and INNO-SPEC- are already camera suppliers of ours, and indeed HYPERA® affords them with an opportunity to increase their sales and turnover by acting as component suppliers (HSI cameras) to our HYPERA® solution (whereby depending on the application, the most appropriate camera will be selected). This could also be the case for and Norsk Elektro Optikk A/S. To this end, these European players in the HSI equipment field are all potential collaborators of ours, whereby their business models are based on equipment/device commercialisation, and they do not specialise in the integration and

[34] Promoting an EU industrial policy for food and drink. Competitiveness Report 2013-2014. Food and Drink Europe.
[35] Eye on China by Ted Agres. Food Quality & Safety magazine, October/November 2011
[36] Paula Hollywood (2012). Using Your Laboratory to Protect Your Brand. ARC View. October 12th, 2012. ARC Advisory Group.
[37] Ensuring Food Safety and Brand Protection Through Supply Chain Traceability. White Paper. CDC Software. Available at: http://www.foodprocessing.com/assets/wp_downloads/pdf/FoodSafetyBrandProtect2009.pdf
[38] Freedonia Group. World Food Processing Machinery to 2016 - Demand and Sales Forecasts, Market Share, Market Size, Market Leaders

customisation of turnkey HSI solutions in the industry (which requires chemometrics, calibration and engineering/integration into the processing lines, as well as after-sales maintenance and servicing). The HYPERA® business model is based on a full turnkey solution that is commercialised based on a core platform technology that is specifically designed for real-time food inspection applications. Our positioning in this `integration gap´ (as discussed in more detail Section 1), heightens the impacts for Europe, as the HYPERA® solution will serve to open up applications for HSI in the global food industry (and pharmaceuticals in quick succession), which will increase demand for European manufactured HSI cameras/imagers. On the other side of the Atlantic, providers of HSI such as CytoViva Inc. offer equipment targeted at research labs that cannot be scaled up for in-line monitoring. Headwall Photonics is a supplier of the Hyperspec® equipment in the UV to SWIR ranges and also covering the midwave infrared (MWIR) and longwave infrared (LWIR). Like SPECIM and INNOSPEC for example, they do not develop integrated solutions. However, a number of companies are positioning themselves as solutions providers, with RESONON from the US and PPO from Canada being the most noteworthy in the context of industrial applications. The unique features of the HYPERA® system in terms of high processing speed, penetration, chemical differentiation, minimal distortion and affordable price allow us to position ourselves against these companies with a very favourable price/performance ratio. Importantly, the existence of emerging competition is a healthy indicator of the existence of a very promising market opportunity. It is ideal timing for a European offering to step into the arena to dominate the European market and leverage a strong and early foothold in the global market.

TABLE 6- COMPARATIVE OVERVIEW OF COMMERCIALLY AVAILABLE HYPERSPECTRAL IMAGING EQUIPMENT

Manufacturer	Country	HSI product/brand	Spectral range	Applications
CytoViva, Inc.	US	CytoViva	Visible near infrared spectral range (VNIR)	Targeted at research laboratories
Headwall Photonics	US	Hyperspec®	VIS, VNIR, SWIR and Extended-SWIR	Targeted at research, industrial applications and airborne platforms
Resonon	US	Pika Ili	400- 900 nm VNIR	Targeted to harsh industrial environments and airborne platforms
		Pika XC	High-performance 400 to 1000 nm	Core device for developing applications in VNIR range
		Pika NIR	900 to 1700 nm NIR HSI	Core device for developing applications in SWIR range
Specim	Finland	AISA	VIS, VNIR, SWIR, MWIR and LWIR	Airborne and defence usage
		SisuROCK	VIS, VNIR, SWIR, MWIR and LWIR	Drill core mineral mapping technology
		SisuCHEMA Breeze	400-1000; 900-1700; 1000-2500 nm	Desktop pushbroom system for chemical and pharmaceutical samples
Norsk Elektro Optikk A/S	Norway	HySpex hyperspectral cameras	VNIR from 400 to 1000 nm and SWIR from 900 to 2500 nm	Airborne platforms, research labs, industry.
P&P Optica, Inc.	Canada	PPO SWIR	850- 2500 nm	Industry applications: Tissue Differentiation; Standoff Detection; Aerial Surveying; Environmental Testing; Quality Control
INNO-SPEC	Germany	Spectral imaging system NIR 1.7	SWIR (900-1700 nm)	Core instrument designed for being integrated in industrial applications
		Color combination NIR 1.7	RGB (VIS) + SWIR (900-1700 nm)	Core instrument designed for being integrated in industrial applications
IRIS	Spain	HYPERA®	SWIR-HSI (900-1700 nm) NIR-MSI (700-1050 nm)	Integrated solution for industrial applications

Regarding a price analysis, first of all the table above describes the principal available cameras; none of them come close to a complex and complete solution such as HYPERA. There are some very simple multispectral VIS-NIR cameras available in the market for as little as €7k, right up to cameras at €80k (high end version-640 pixels), which do not offer software or integration in the processing line. Our HYPERA combination of 2 cameras (VIS-NIR and SWIR), lighting, software, housing for harsh environment and for in-line integration, at €80k is very well priced (this prices will be validated during the Phase 2).

MOST RELEVANT MARKET SEGMENTS FOR INTRODUCTION OF THE NEW SOLUTION

We are initially targeting food processing applications (and once we have secured traction in this sector, we will commence introduction into the pharmaceutical sector). The HYPERA will be fully deployed in the 'first client' group (CASA TARRADELLAS, PLATOS TRADICIONALES, MARÍN GIMENEZ, DAWN FARM FOODS) during the Phase 2. These companies will serve as powerful endorsers of the HYPERA® in the pizza, meat, fruit and vegetable processing and RTE foods segments. In the commercialisation phase, we will leverage application notes from these business cases to build sales in Europe and globally targeting these and other segments where there are clear technological gaps and increasing industry interest in investing in solutions for automatic quality control, and for applications in which HSI has already shown considerable achievements in addressing quality and safety issues in previous studies: fish and other seafood[39],[40], grains, dairy products, and horticultural products[41],[42],[43],[44], among others. Once we have gained traction and sales in the food industry we will target the pharmaceutical industry (planned for 2021).

[39] ElMasry G, Iqbal A, Sun D-W, Allen P, Ward P. 2011a. Quality classification of cooked, sliced turkey hams using NIR hyperspectral imaging system. J Food Eng 103:333–44.
[40] ElMasry G, Sun D-W, Allen P. 2011b. Non-destructive determination of water-holding capacity in fresh beef by using NIR hyperspectral imaging. Food Res Int 44:2624–33

MARKET BARRIERS TO BE OVERCOME

The major barriers preventing uptake in the **food industry are cost, speed and accuracy**- HYPERA is overcoming each. Food legislation that places food quality and safety responsibility with the producer will be a driver for adoption of HYPERA and evidence based application notes and publications from the project will be powerful documents for demonstrating the accuracy and reliability of the HYPERA system for food quality inspection. A solid approach is to first build the HYPERA brand with strong industry case studies and establish a client base with international names from the food industry, and later progress to enter the pharma sector, whereby the most significant barriers to market entry will be regulatory and validation requirements (e.g. CFR 21 11).

TARGET USERS AND HOW THEY WILL BE REACHED

The targeted users of the final HYPERA solution are food companies (and pharmaceutical companies in the post-project phase) both SMEs and large enterprises. This is a global market. Our initial effort will focus on targeting the **headquarters of large-size companies in the Food and Pharma industry**, which have been defined as the early innovator group, due to a) their high production volumes justifying the cost of an automatic food inspection system; the probability of follow-up contracts is high for large-size companies: once first in-line monitoring units are successfully tested, the probability is high that they ask for further units to roll-out automatic food quality inspection in their international production plants. IRIS will contact directly with headquarters because they are usually in charge of implementing new technologies and process improvements. Headquarter (HQ) locations of the Top100 companies in the food[45], pharma[46], are primarily located in the EU (mainly Germany, Switzerland, France, UK and Italy), North America (US and Canada) and Asia (mainly Japan and India, and China). Selling the HYPERA® product involves selling the equipment, the system calibration (acquiring spectra of client samples, conducting laboratory referential analysis and elaboration of predictive models) and integration in the production line (engineering). Due to a relative high service proportion, the direct contact with important clients will be crucial. For this purpose, IRIS has plans to open regional offices to acquire high-priority clients, establish agreements with distributors, manage distributors and deliver the service part during and after the acquisition process (in 2015 we opened an office in New York and San Francisco). Distribution roll-out will take place in two phases as is described below under the heading *'Strategy plan for commercialisation of the business innovation project'*. IRIS has an experienced Sales and Marketing Department and the experience to effectively reach the decision makers (Plant Managers, Quality Managers, Production Managers, R&D and Innovation Directors) in the process industries is present. Targeted communication campaigns to companies from IRIS existing client-base and networks (which include such large industries as DAIRYGOLD, NESTLE, DAWN MEATS, GSK, SANOFI, ALMIRALL, among others) and beyond will be designed (via electronic mailings, paper mailings, personal/face-to-face communication…), and through the channels of our distribution/sales agents. Prospective customers will be found using a combination of a website with application examples and technical presentations at conferences and exhibitions (listed under 'Communication' below).

2.2 MEASURES TO MAXIMISE IMPACT

A) DISSEMINATION AND EXPLOITATION OF RESULTS

KEY STAKEHOLDER INVOLVEMENT FOR SUCCESSFUL EXPLOITATION

The following stakeholders have been identified for their possibility to add value and to facilitate the commercial exploitation of the HYPERA®: ■ **Additional beta clients** from different food segments, pharmaceutical processing (and chemical industries) who will enable us to create additional application studies and Business User Cases; ■ **Reputable and effective distributors and sales agents** for building our European and global commercial network, along with **OEMs** that are interested in integrating HYPERA into their equipment to create a 'smart' or 'intelligent' equipment offering; ■ **Industry Associations and Clusters** who can provide a very powerful pathway to their members. They also enable active engagement with policy makers and in helping the industry and policy-makers to work together towards common goals in relation to process efficiency, quality, safety, sustainability, etc. ■ **Spectral imager providers**, whereby HYPERA® provides them with an integration partner to mainstream the adoption of their sensors in in-process industry applications. ■ **Equipment certification consultants** to facilitate all the steps involved in obtaining certifications; ■ **Renowned academics and industry influencers** who can independently test the HYPERA system and publish/disseminate the results.

FURTHER STEPS NEEDED BEFORE THE RESULTS ARE MARKET READY

Our approach in relation to planning the steps towards being market ready is centred on the well-established **4 Ps Marketing Mix- Product, Price, Promotion, Place-** (Figure 4). Based on past experience we know this is a robust and complete approach to planning a pathway from product to market. Central to the methodology is the **target market** (discussed above), so that we always keep our target market (i.e. the food industry- and eventually the pharma market- and more specifically the decision makers within this industry,

[41] Karimi Y, 2012. Application of hyperspectral technique for color classification avocados subjected to different treatments. Food Bioproc Technol 5:252–64.
[42] Cen H, Lu R, Ariana DP, Mendoza F. 2013. Hyperspectral imaging-based classification and wavebands selection for internal defect detection of pickling cucumbers. Food Bioproc Technol 7:1689–1700.
[43] Wei X, Liu F, Qiu Z, Shao Y, He Y. 2013. Ripeness classification of astringent persimmon using hyperspectral imaging technique. Food Bioproc Technol 7:1371–80.
[44] Yu K, Zhao Y, Li X, Shao Y, Zhu F, He Y. 2014. Identification of crack features in fresh jujube using Vis/NIR hyperspectral imaging combined with image processing. Comput Electron Agric 103:1–10.
[45] http://www.foodengineeringmag.com/global-top-100-food-&-beverage-companies
[46] https://www.google.com/fusiontables/DataSource?docid=1uEZuNGJlP68Imym7zAVT-XHXwhHdvs2lT5L6iNs#rows:id=1

who will decide to invest or not in our HYPERA solution). Each of the ´Ps´ is discussed below in the context of HYPERA and our target market.

PRODUCT- As we are dealing with industrial inspection equipment, our product design, features and functionality are the most important factors. During the Phase 2, we will structure the value proposition of our new product (to validate and improve the value propositions and USPs we have already formulated- see Figure 3 above). We will draw up the product datasheets that will outline the key information, and we will also create product brochures outlining applications (the applications that have been validated in this project and additional applications that we plan to quickly serve in the commercialisation phase), highlighting the benefits, etc. The services that support the product will also be clearly defined (maintenance service, customisation services, upgrades to the software and models), consultancy based on system use and data understanding. We will also further build the HYPERA brand identify. We expect all Product-related activities will be completed by the end of the Phase 2.

PRICE- In the Phase 2 we will validate and finalise our pricing based on an up to date knowledge of costs in line with developments, our pricing policy in relation to discounts, credit terms, etc. We have initially set price for an installed unit at € 80k based on our existing technology and market experience. During the Project, we will finalise and validate a price list based on our manufacturing and logistics costs, our required margin to have a profitable business, the necessary margin for distributors/sales agents, and our necessary pricing positioning vis-à-vis market tolerance and competitors.

PROMOTION- Measures consist of the communication activities we will use in the marketplace to increase awareness about the product and its benefits to the target segment. Our post project campaign (the communication campaign during the action is discussed in Section 2c below) will consist of: ■ Official product launch, along with a press conference; ■ Exhibiting at shows; ■ Presentations at conferences ■ We know from past experience that the strongest promotional tool is successful applications. As such, building the development round early adopters gives the product the evidence to prove capability and the push to have strong growth from day one. Publishing articles in industrial/sectoral channels (technical/sectoral magazines, etc.), publishing whitepapers and additional application notes on the product page of our website, etc. ■ Advertising via digital ad campaigns; ■ Featuring as sponsors at a number of key industry focussed conferences/events. ■ PR campaigns whereby we will aim to reach out to industry influencers who could be a powerful source of reviews ■ Authoring feature blogs as guest bloggers in on-line sources; ■ Our sales staff will also play a major role in promotion of the product, and in the Phase 2 they will be trained fully on the HYPERA ■ Sales promotion: we will use this tool for example for our launch campaign, whereby companies that agree to buy during the launch period will benefit from a discount, we will also have sales promotions linked to exhibitions, etc.

FIGURE 4- THE '4 PS' MARKETING MIX

Finally and very importantly, **PLACE**…in other words, how to get to our product to a global market. A key requirement is to establish a manufacturing capability to deliver the product. During the Phase 2 project we will establish our manufacturing and associated logistics (Task 4.4), by the end of which we will have produced and quality approved a final product unit. In the post project phase we will build additional stocks and we will also set up our warehousing inventory. In relation to how we plan to serve the global market, as is discussed below, we will have our own trained Sales Team (supported by an Application Team) and we will set up our Distribution networks in two stages (as is described under the next heading). We will also establish agreements with shipping companies for transportation and insurance companies. Through *WP5-Gear up to market*, we will implement, build on and further detail plan this initial 'Plan for the Dissemination and Exploitation of the Action's Results'.

STRATEGY PLAN FOR COMMERCIALISATION OF THE BUSINESS INNOVATION PROJECT

In relation to our **commercial strategy**, the HYPERA will be commercialised by IRIS. In-situ inspection systems have an extended sales pipeline, with considerable upfront customer interaction, evaluation and testing, and due to the nature of the process manufacturing industries, process analytical systems require some level of customisation, and they also need to be integrated (and especially as we are offering a turnkey solution) and ongoing support has to be ensured. As such, there are two key aspects of the business plan; the creation a **specialist Application Team** and the upfront development of a **Network of Agents and Distributors;** to support the sales and deployment activities. The process will be driven by the core **Sales Team** that will be led by Mrs. Johanna Granda, the Commercial Director at IRIS. Marketing process analysers and inspection systems is a specialist business, our **Application Team** will be made up of dedicated engineers who will provide support to the sales team, such as testing materials and

detailed site visits, advice and support during system installation, provide training and to handle both site and remote the customer support.

In order to benefit from local knowledge and experience of selling process systems, as part of the project we will build a network of **agents and distributors to provide a point of contact for customers and follow up on the sales process**. As such, our international commercialization strategy will be to start with an agency/distributor network for prospecting and opportunity validation in the key regions not reached by IRIS for marketing and selling the product supported by the in house Application Team. We see these relationships being long term partnerships and ultimately growing these into regional centres of expertise able to deliver complete solutions. Hence over time these regional centres will develop local application teams and the local sales support will be complimented by the establishment **regional integration agreements with engineering companies** to directly support our commercial and technical centres. Agreements will also be established with global OEMs for delivering built-in HYPERA systems in their equipment (e.g. METALQUIMIA, GLATT, GEA, etc.).

In relation to our **current and planned geographical presence** (see figure 6 in the IRIS participant description in Section 4.1), IRIS currently has production and commercial facilities in Ireland (which allows us to serve UK as well) and Spain (which enables us to serve Portugal and Italy), and as part of our further internationalisation strategy we have recently opened technical/sales offices in US East Coast (New York City) and West Coast (San Francisco) and in 2016/2017 we will open an office in Germany (that will also serve Switzerland and Austria), France (which will serve Benelux), Japan (for our Asian presence) and Mexico.

Time to first sales is Jan. 2019, when we will commence the sale of first units to our beta clients in the immediate post-project phase and we will build out ours sales effort from there. The end-user involvement in the Phase 2 of this Innovation Project, through our 'first client' or early adopters from the food industry will provide a quick route to first sales if industry trials at each site are successful.

B) INTELLECTUAL PROPERTY, KNOWLEDGE PROTECTION AND REGULATORY ISSUES

INDUSTRIAL PROPERTY RIGHTS ASSETS

IRIS owns the HYPERA® brand and all the know-how behind the HYPERA® development to date in relation to the TRL6/7 development of the novel combination of Shortwave Infrared Hyperspectral Imaging (SWIR-HSI) in reflectance with Near Infrared Multispectral Imaging (NIR-MSI) in reflectance/transmittance, along with the copyright of the chemometric software, and the results of tests for the detection of foreign bodies in foodstuff whereby the SWIR-HSI/NIR-MSI and chemometrics enables a) penetration (NIR), b) composition differentiation (SWIR); and c) minimising the image distortion caused by scattering (chemometrics). IRIS will **own all the knowledge items that will be generated through this SME Instrument**. In relation to our freedom to operate, we have a patent reviews (discussed below) that concludes that no existing patent directly competes with the envisaged HYPERA ® system.

MEASURES FOR ENSURING COMMERCIAL EXPLOITATION

A systematic Technology Watch task will be implemented over the course of the work plan (through T6.1) to ensure we keep up to date of relevant competing developments. In addition, every 6 months a patent review will be performed to reconfirm our 'freedom to operate' and to ensure we are not infringing on the IPR of a third party. A SWOT analysis will be prepared at M6 of the project by the IRIS Management Board and it will be reviewed and updated each quarter in relation to the HYPERA ® solution strengths and weaknesses and in the context of the market opportunity and any threats to realising commercial exploitation. A major effort we have made in order to pave the way forward for commercial exploitation has been lining up a powerful group of "early adopters", through our 4 first beta client companies of different sizes, representing different segments of the food processing industry (pizza, meat, fruit and veg processing and RTE foods). These will allow us to gain serious endorsement value from the Phase 2 project and to document impactful case studies/business cases in the use of HYPERA® for automatic food inspection applications.

KNOWLEDGE MANAGEMENT AND PROTECTION STRATEGY

A patent search has been performed to confirm our freedom to operate. The last patent search was performed on January 25th, 2016, using the World Intellectual Property Organisation (WIPO) PATENTSCOPE database (https://patentscope.wipo.int/search/en/search.jsf) using the key words: ALLTEXT: "hyperspectral" + "foreign bodies"; ALTEXT: "chemical imaging" + "foreign bodies"; ALLTXT: "multispectral + "foreign bodies" The results revealed that no patent exists that would affect the HYPERA development. In fact, many of them describe the optical setup of the core device while others are associated to specific targets (for example, fruits or insects) while HYPERA is based on adapting off-the-shelf core devices and is intended for generic applications in detection of foodstuff-borne foreign bodies no matter the specific nature of the product to be monitored. In this set of patents the following are particularly relevant: "Method and system for contaminant detection during food processing" (US6587575B1, Georgia University), which is limited to use a few wavelengths the VIS-NIR range, very basic algorithms and aimed at detecting fecal and ingesta on meat and poultry carcasses; "Non-destructive detection method and device for agricultural and animal products based on hyperspectral image technology" (CN1995987A, chinese institution), which is limited to the VIS-NIR range, very basic algorithms and aimed at characterizing agricultural and livestock products; "Equipment and method for analyzing image data" (EP2343671A2, Sumitomo Electric Industries, Ltd.), which is limited to using a variety of the SVM algorithm for processing

hyperspectral data; "High spectrum image collector for fruit" (CN2874482Y, Chinese institution), which merely describes fruit hyperspectral image acquisition device without mentioning spectral range or data processing tools; "System for multi- and hyperspectral imaging" (US7835002B2, RP Ventures Tech Office AB), which is focused in the optical setup for acquiring the hypercube without any mention to generic or specific application, lighting or data processing; "Imaging spectrometer" (US8994939B2, SPECIM SPECTRAL IMAGING Ltd Oy), which is focused in the optical setup for acquiring the hypercube without any mention to generic or specific application, lighting or data processing; and " METHODS, SYSTEMS AND DEVICES FOR DETECTING INSECTS AND OTHER PESTS" (WO/2013/017860, UNIVERSITY OF EAST ANGLIA), which is not based on IR spectroscopy but in Raman and fluorescence excited at 600 nm. Basically, we can conclude that there are no evident IP conflicts between HYPERA and existing (patented) similar systems because the most of the related patents are constrained to specific applications or specific sensing technologies such as VIS-NIR hyperspectral imaging.

In relation to the novelty of our HYPERA development, in line with the design and development work that will be performed in the Phase 2 we will analyse all outputs for their novelty and for their protectability through Patents, Trademark, Industrial design, Copyright, Trade-secret, or Confidentiality, as appropriate (i.e. the form which offers the most adequate and effective protection for each type of result). We envisage that we will gain IPRs through a number of aspects of the development (all of which will be confirmed and pursued as the developments unfold). In relation to the **patentable results** of our development, it is expected that the following elements of our developments will be patentable: The specific parameters (settings) for the combination of Shortwave Infrared Hyperspectral Imaging (SWIR-HSI) in reflectance with Near Infrared Multispectral Imaging (NIR-MSI) in reflectance/transmittance for industrially reliable foreign body detection and chemical monitoring of materials. Moreover, we will study the feasible protection routes for the combination of chemometrics (patent possibilities for US, and copyright or other in Europe). As addressing the ideation of such solutions will be tackled according to the TRIZ methodology, it is envisaged that some of the outputs could be novel and specific enough for being patented. In fact, one of the advantages of using such a methodology is its productivity in terms of novelties. We will also submit a trademark application for protecting the HYPERA brand name and its logo.

REGULATORY AND/OR STANDARD REQUIREMENTS TO BE FULFILLED

All EC and national regulations that could influence on the commercialisation of the project outputs will be studied and their influence on the development or the future use of HYPERA product will be taken into account. The requirements in relation to adherence to: ASTM E1655-97; ASTM E1866-97; ASTM E1790-04; IUPAC Guidelines for Calibration in Analytical Chemistry; EMEA/CHMP/CVMP/QWP/17760/2009 Rev2; 11 CFR Part 35; 21 CFR Part 11, EC Machinery Directive and food-processing equipment (EU 98/37/EC Directive), etc.) and standards (e.g. EN45011 or ISO65 quality standards and all manufacturing standards set by ISO and to the GMP regulations, as well as to compliance with the CE Mark, will be given particular attention during the design stages of the system. ATEX certification requirements will also be fulfilled. Through Task 1.3, the regulatory and standard requirements will be studied with the end-users to ensure that all necessary actions that need to be performed during the development work in Phase 2 are carefully planned. During Task 4.1, certification applications will be made and the appropriate steps will be initiated towards obtaining approvals by the standards bodies and to ensure compliance with guidelines, standards and regulations so that HYPERA® can be sold and used in food and pharmaceutical processing in Europe and internationally. It is expected that regulations in terms of food safety will be drivers for adoption, e.g. Regulation (EC) No. 178, for example, establishes the European Food Safety Authority and outlines the Global Food Law (GFL), focusing on traceability, product withdrawal, recall, and notification for food and feed related to safety requirements, including imports and exports, very similar to the Bioterrorism Act outlined by the U.S.

MEASURES TO PROVIDE OPEN ACCESS TO PEER-REVIEWED SCIENTIFIC PUBLICATIONS

It is our intention to publish aspects of the project results in peer-reviewed scientific publications, and these will be made available free of charge through Green Open access publishing.

C) COMMUNICATION

In line with our 4Ps Marketing Mix, the communication activities that will be performed over the course of the project will have the principal objective of **communicating the benefits and value of our HYPERA product to our future clients**. Initial communication activities within the food (and pharma) industries at exhibitions, have highlighted the need for and interest in our solution and we believe that the industry will be highly receptive to this message. Early testing provides material for early promotion. A such, during the Phase 2, we intend to build the brand identity and promote the product actively through standard means such as the project web, leaflet, poster, etc., as well as through more vanguard and dynamic tools such as blogs, social media (LinkedIn and twitter), Infographics, etc. We will also produce white papers as a very powerful and impactful communication tool. Digital marketing will also be employed for reaching users. In Internet marketing, storytelling is a potent method of converting sales prospects to customers. We will use the 'business stories' technique, whereby we will document case studies from our early adopters (no sensitive processing information will be disclosed and only relevant non-confidential information will be used) to communicate the business benefits of investing in the HYPERA system in an impactful, creative and engaging way in order to effectively connect with industry decision

makers. We will showcase and exhibit results at 4 exhibitions/tradeshows over the course of the project (e.g. Alimentaria (Barcelona), FoodTech- formerly BTA- (Barcelona), ANUGA (Cologne), ANALYTICA, etc.) and in at least 4 conferences (e.g. Global Food Safety Conference, Annual European Food Manufacturing & Safety Summit, Pharma Process (Barcelona), APACT, EuPAT, etc.). We will also publish 4-6 articles in industrial/sectoral channels (technical/sectoral magazines, etc.). Publications of non-confidential and protect results in peer reviewed journals (e.g. J. of Spectroscopy, J. of Food Processing, J. of Process Control, MORE, etc.) are considered as very powerful marketing tools to the industries, such as food, pharma and other sectors of interest (such as medical), as well as to policy makers and standards bodies, and we will aim for at least 2-3 papers resulting from the project. We will mobilise our membership in networks and cluster memberships to promote the HYPERA solution and to participate in events/workshop, etc. that would enable us to present HYPERA and disseminate the successful case studies: SPIE (International Society for Optics and Photonics), A. SPIRE (Sustainable Process Industries), SECPHO (Southern European Photonics Cluster), among others. In short, our communication activities will ensure we maximise the future dissemination and exploitation of the project results. Finally, as we are benefitting from tax-payers contributions, there is scope through social media to make the general public aware of the HYPERA development and the associated societal benefits, e.g. contribution to improved industrial competitiveness, job creation, prosperous regions, access to high quality and safe foods (and pharmaceuticals); these are all benefits that contribute to our quality of life as EU citizens. We will benefit from highly qualified digital marketing professionals that form part of the marketing team in IRIS- whereby they have designed numerous successful digital media campaigns in the past, which is a powerful channel for reaching increasing segments of the population (including industry) through Social Media communication.

3. IMPLEMENTATION

3.1 WORK PLAN – WORK PACKAGES, DELIVERABLES AND MILESTONES

OVERALL STRUCTURE OF THE WORK PLAN

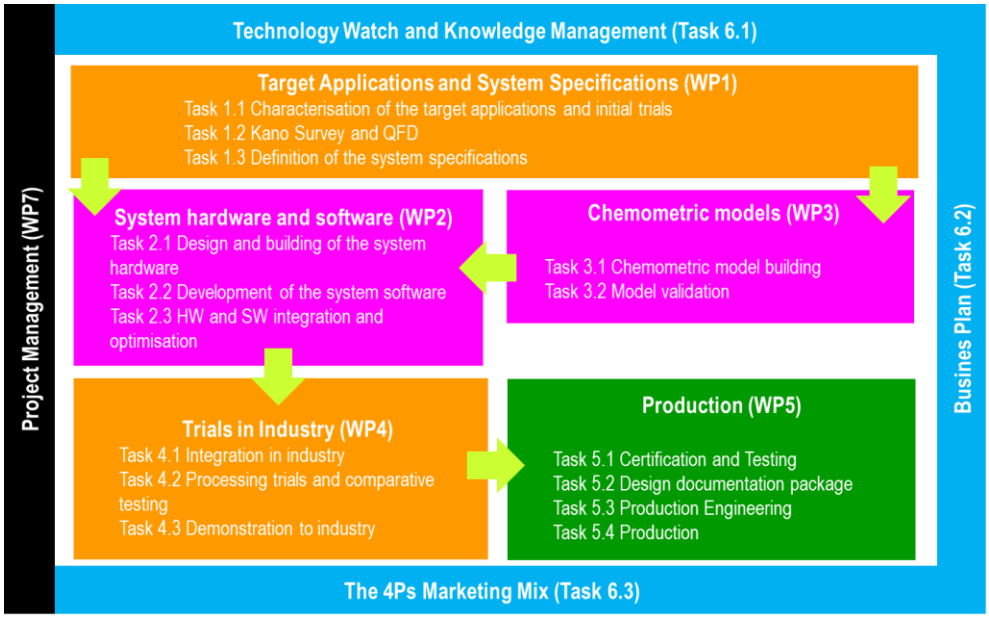
We have built up our work plan based on our approach covering 3 project phases of a) application, b) product development and c) commercialisation planning. It is 24 months in duration and consists of 7 work packages (WPs) as follows: **WP1** will commence by gaining a detailed understanding and characterising the selected process applications at CASA TARRADELLAS, PLATOS TRADICIONALES, MARÍN GIMENEZ, DAWN FARM FOODS. Using the prototype system, we will undertake an initial demonstration of the system on these applications to provide test data for the requirement specification and development project. In **WP2**, the chemometric models will be developed and validated using the prototype systems and client samples from WP1. Optimised models will be implemented in the system software in WP3. In **WP3,** the imaging systems will be optimised for the selected applications and a robust industrial hardware design will be performed and the system software platform will be developed (and the chemometrics will be implemented) and integrated with the suitable enclosure and safety system for the application. In **WP4**, the integrated system will then be alpha tested on the selected applications at the end user sites by the end users and supported by the development teams to validate the design. Having validated the design we will translate the prototype to a product (**WP5**) through the creation of the manufacturing documentation package using the engineering information and specifications. This will be passed to purchasing and production engineering to enable manufacture. Deployment of these first products will be in our participating end-users and to additional selected Beta customers to enable early refinement and troubleshooting support by the Application Team. In parallel with the product development work, we will continually build on and upgrade our business development plan (**WP6**). As part of this activity the marketing plan will review the business opportunities of the technology and devise a sales strategy (building on our initial strategy contained in our business plan). A project management framework will be implemented through **WP7**. The Gantt Chart (Below) and Pert diagram (at end of this section) show the timings and the interdependencies between the different WPs and tasks and the 1-page Gantt (which also includes the deliverables and milestones) is a very effective tool for high-level project control and helicopter overview for effective project management.

TIMING OF THE DIFFERENT WORK PACKAGES AND THEIR COMPONENTS

No	WPs and Tasks	Del. & Milestones	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
1	Target Applications and System Specifications	ML1			#																					
1.1	Characterisation of the target applications and initial trials	D1.1		■																						
1.2	Kano Survey and Quality Function deployment (QFD)																									
1.3	Definition of the system specifications	D1.2			■																					
2	System Hardware and Software	ML2, 4, 5, 8						#				#		#							#					
2.1	Design and building of the system hardware instrumentation	D3.1, D3.2				■		■																		
2.2	Development of the system software	D3.3, D3.4										■		■												
2.3	Integration of HW and SW, functionality testing and optimisation work	D3.5												■												
3	Chemometric models	ML3									#															
3.1	Chemometric model building	D3.1, D3.2							■																	
3.2	Model validation	D3.3							■		■															
4	Trials in industry	ML6, ML7, ML9														#				#				#		
4.1	Installation in industy	D4.1														■										
4.2	Processing trials and comparative testing	D4.2, D4.3																		■				■		
4.3	Demonstration to food processors	D4.4																								■
5	Production	ML8, ML11																			#				#	
5.1	Certification and Testing	D5.4																								■
5.2	Design documentation package	D5.1																			■					
5.3	Production Engineering	D5.2																			■					
5.4	Production	D5.3																							■	
6	Gear up to market	ML10																						#		
6.1	Technology Watch	D6.3																		■						
6.2	Knowledge management	D6.4																						■		
6.3	Communication, dissemination and training activities	D6.1, 6.2, 6.5			■									■												■
7	Project Management																									
7.1	Monitoring and reporting	D7.1, D7.2																								
7.2	Quality and Risk Management																									
7.3	Innovation Management		■		●			●			●			●			●			●			●			●
7.4	Fulfilment of Contractual Requirements		◊		■			■			■			■			■			■			■			■

■ Deliverable / # Milestone / ◊ Project Kick Off Meeting ● Meetings of the IRIS Management Board

GRAPHICAL PRESENTATION OF THE COMPONENTS SHOWING HOW THEY INTER-RELATE



3.2 MANAGEMENT STRUCTURE AND PROCEDURES (ONLY TO THE EXTENT RELEVANT IN SINGLE ENTITY PROPOSALS)

As this is a single entity project, the organisational structure (Figure 5) is practical and simple, yet highly effective. The **Project Manager (PM) for the project is Dr. Edurne Gaston.** She is the leader of the Food Technology group in IRIS and a highly experienced project manager. She also has strong experience in researching the use of HSI technology in food processing applications. For the HYPERA® Innovation project, Edurne will be responsible for ensuring that the work programme adheres to its schedule, fulfils its objectives (according to the principles of on-time, on-budget and to high quality), and that the project meets the requirements of the company. The Project Milestones (Table 3.2 a) will be used as a valuable tool for determining whether or not the project is on schedule. She will be responsible for the management of the internal project development team, as well as in managing the relationship with stakeholders (early adopters of the HYPERA® system from the IRIS existing client base) who will be involved at different stages of the work plan. She will be active in steering the evolution of the project by closely monitoring and controlling the progress of the work, as well as the engagements and contributions from the stakeholders. She will be responsible for ensuring that the project is managed in adherence with the internal project management procedures that are certified by ISO 9001:2008 and UNE-EN-ISO 166002: Management Systems for R&D&I, and with support from the IRIS Project Management Office (PMO). Edurne will also be the interface with the sponsors (the European Commission) on behalf of IRIS, and she will ensure that all contractual obligations are fulfilled. Edurne's CV in Section 4.1 highlights her proven experience in managing collaborative and B2B R&D projects.

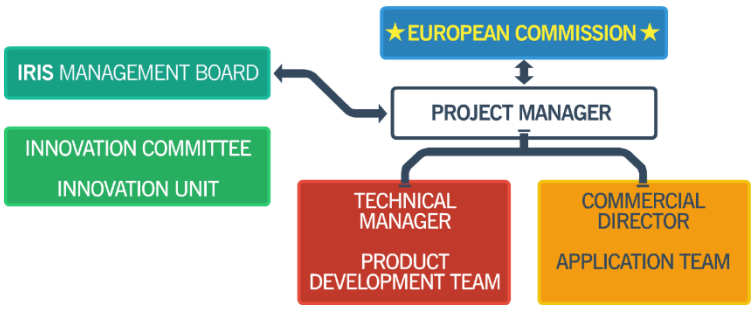


FIGURE 5- MANAGEMENT/ORGANISATIONAL

The **IRIS Management Board, Chaired by General Manager, Mr. Colm Digby,** will be responsible for the strategic management of the project in line with the business and commercial strategy of the company. The IRIS Management Board will take all strategic decisions in relation to the project (e.g. decisions with regard to any major technical decisions, budgetary decisions, knowledge management/IP protection. Decisions will be taken following the normal decision making procedures and voting rights of the IRIS Management Board. The PM will formally report the progress of the project to the Management Board every month and every quarter the budget spend will be reported to the board. Extraordinary meetings with the Management Board will be called as required to address, discuss and arrive at a swift resolution should any extraordinary problems or issues arise. Mr. David Montes, the Technical Manager in IRIS, will lead the IRIS Product Development Team that will implement the work according to the project work plan. He will meet with the Project Manager and the development team on a weekly basis for steering technical discussions, monitoring the evolution of the work, measuring progress against the work plan, analysing potential risks and bottlenecks, contingency planning, etc. Issues that could have a strategic or critical impact on the project will be reported immediately to the General Manager, who will convoke an extraordinary meeting of the Management Board as appropriate. Mrs. Johanna Granda, the Commercial Director at IRIS, will lead the **Commercialisation team**, and she will be responsible for building the sales agent/distribution network internationally as well as further developing the business plan (with the inputs and approval of the Management Board). She will also coordinate with the marketing team to ensure the project dissemination and communication campaigns are aligned with the commercial strategy and she will lead the project's engagement with the 'first clients'/early adopter end-users (CASA TARRADELLAS, PLATOS TRADICIONALES, MARÍN GIMENEZ). Together the Technical Manager, the Commercial Director will establish the Application Team that will provide technical support to the sales team.

■APPROPRIATENESS TO THE COMPLEXITY AND SCALE OF THE PROJECT: The HYPERA® project will be managed following the same structure and decision making mechanisms as an internal product development project within IRIS, and this is considered as a highly appropriate approach. ■INNOVATION MANAGEMENT: Innovation Management is a function that exists within the IRIS organisation structure, and is part of our 'UNE-EN-ISO 166002: Management Systems for R&D&I' certification. The structure consists of an **Innovation Committee** (comprised of the General Manager, Director, Deputy Director and the Commercial Director), and an **Innovation Unit** (which is led by the Innovation Unit Head- a role that is occupied by the Science and Technology Manager in IRIS, along with senior scientists, researchers and engineers from the company). The Innovation Committee (which is an advisory body to the Management Board) meets with the Innovation Unit on a monthly basis in order to track the innovation potential of all active projects, to analyse IP, to discuss technology watch, state-of-the-art, competition, and market considerations around the R&D lines that the company is actively investing in. The Innovation Unit also assist the Innovation Committee in the development, maintenance and updating of the R&D Roadmap and the technology portfolio. The innovation management of HYPERA will be performed by these Innovation Bodies within IRIS. ■CRITICAL RISKS: The critical risks (technology-, commercial-, and organisation-related) that could jeopardise this Innovation Project and appropriate risk-mitigation

measures are provided in Table 3.2b. Through Task 8.2, a continual process of risk assessment and monitoring will ensure that the status of all risks is effectively managed and that action plans for all mitigation measures and contingency plans are implemented, monitored and reported. The process of risk identification and management will be based upon the principles and guidelines of the FERMA (ISO 31000 related) Standard[47].

[47] Federation of European Risk Management Associations (www.ferma.eu)

DAWN FARMS
GREAT INGREDIENTS, GREAT FOOD.

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E info@dawnfarms.ie

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Dear Oonagh,

I would like to confirm our interest in collaborating with IRIS in your HYPERA development, in the event that you are successful in your SME Instrument application under Horizon 2020, whereby we would be interested in working with you to define the specifications for the system and in testing and validating the system.

As we have discussed we are about to install a new continuous high volume multi species cooked line (stainless steel belt running at speed of 7-10m/minute) and we are looking for an in-line foreign body detection solution for bone & foreign body detection, particularly low density products. To this end, we feel that this would be a very good opportunity to test the hyperspectral imaging solution that you are developing.

Best wishes,

Conor Cahill

Head of Science and Innovation

Dawn Farm Foods

15th June 2015

AIMI TMI foods

Platos Tradicionales

Buñol, 9th June 2015

Dear Ms. Granda,

Thank you for the opportunity to cooperate with you in the HYPERA development that you are submitting to the Horizon 2020 SME Instrument on June 17th 2015.

As discussed, PLATOS TRADICIONALES would be very happy to act as an industry validation site for your hyperspectral imaging foreign body detection solution.

PLATOS TRADICIONALES manufactures ready meals for MERCADONA and is owned by MARTÍNEZ-LORIENTE, the Spanish leading fresh meat processor, and FLEURY-MICHON, the French leader company in the ready meals sector. The PLATOS TRADICIONALES plant in Spain has become one of the most modern factories of manufacture in Europe. The installations of PLATOS TRADICIONALES are provided with the most sophisticated security systems within their employees making sure the consumer gets the highest guarantees and the highest alimentary security.

Efficient and reliable detection of foreign bodies is crucial for PLATOS TRADICIONALES. Currently the plant combines metal detector, X-ray and visible inspection with human eyes to detect foreign bodies. To increase the efficiency of foreign bodies inspection we would be interested the hyperspectral imagining foreign body detection solution. The ability to detect foreign bodies below the surface and to automatize the detection process is highly attractive for us.

Yours sincerely,

José Antonio Mas Pons

General Manager PLATOS TRADICIONALES, S.A.

Dear Johanna,

I would like to confirm our interest in collaborating with **IRIS** in the capacity of an end-user of the **HYPERA** system in the event that you are successful in your SME Instrument application under Horizon 2020.

As you know through our past engagements, **CASA TARRADELLAS** is interested in technology that can offer an affordable and reliable foreign body detection solutions for our fresh pizza production.

We are especially interested in the ability to detect foreign bodies, such as pieces of plastics, carton and wood on pizzas. These foreign bodies cannot be detected with visible cameras because colors are very similar to the pizza topping. Furthermore in some cases foreign bodies are located below the surface and therefore not detectable with visible cameras.

As leading Spanish provider for fresh pizza our daily effort is guaranteeing the maximum quality in our products and the reduction of foreign bodies is one of our strategic goals.

To this end, we are very interested in defining together with you the industrial requirements for the hyperspectral imaging solution you propose in order to ensure that the developed solution corresponds best to industry needs. Furthermore we would provide access to our pizza production as test site/ case study.

We wish you every success with your application, which we feel is a very worthwhile development. We look forward to a positive outcome.

Best wishes,

CASA TARRADELLAS, S.A.
C.I.F.: A 08.800.450
Ctra. Barcelona - Puigcerdà, Km. 70
08503 GURB
Apdo. Correos: 3[illegible]500 VIC (Barcelona)

Josep Terradellas Falgueras
Director R & D
CASA TARRADELLAS, SA

Date: 08/06/2015

MarínGiménez
fruit solutions

Moratalla, 11th June 2015

Dear Ms. Granda,

As discussed in our meeting today, 11th of June, I would like to express our interest to participate in the development of the HYPERA system in the framework of Horizon 2020 SME Instrument, to be submitted on June 17th 2015.

We, Marín Giménez, are a company specialized in **fruit processing for industrial use**. Our main objective is to produce the highest quality **products with the maximum level of safety standards** whilst maintaining competitiveness.

One of our daily challenges is the detection of foreign bodies: Currently we use a visual inspection system and X-Rays, however with these technologies we are not able to detect foreign bodies with low density. Therefore we complement quality inspection with human visible inspection: More than 50 employees are dedicated to ensure highest quality.

For us, the successful development of the HYPERA system would be a breakthrough: An efficient detection of foreign bodies with minimum human intervention would increase significantly our quality and decrease at the same time human intervention costs.

Our goal is to stay competitive. And in a globalized world with competition from China and other emerging markets, the only way to stay competitive is seeking highest quality standards by employing disruptive and cutting-edge technologies.

Therefore **we would be extremely happy to offer our facilities as testing and validation plant**.

Yours sincerely,